

Causal State Binding: An Intervention-Based Evaluation Framework for Language Agents

Xiao Jia

School of Artificial Intelligence, The Chinese University of Hong Kong, Shenzhen

xiaojia@link.cuhk.edu.cn

ORCID: <https://orcid.org/0009-0006-7711-4067>

Supplementary Overview

This Supplementary Information reports the dataset-resolved evidence, metric definitions and reproducibility details for the causal state-binding study. The main text introduces reader-facing names such as causal state binding, structured agent condition, decisive intervention state, trace-action coupling and hidden-target finite-action probes; the alphanumeric labels in this supplement are raw protocol identifiers retained for exact export matching. The tables cover the implemented agent family, prompt/event corpora, finite-action-code behavioral benchmark, open-weight no-context and strict-lesion validation, decisive-field positive controls, local control-binding wrapper checks, local forced-budget entropy/token matching, a SWE-bench Lite issue-to-file localization validation with reliability calibration and threshold analyses, matched-interface action-field coupling controls, baseline lesion matrix, stochastic and entropy-calibration analyses, API evidence, blinded audit and interface diagnostics. Values shown in the PDF tables are rounded for readability; the CSV exports retain the full exported precision. Entropy summaries, resampling intervals and agreement statistics follow standard empirical entropy, bootstrap and nominal-agreement conventions (Shannon, 1948; Efron and Tibshirani, 1993; Cohen, 1960).

Extended Data Figures

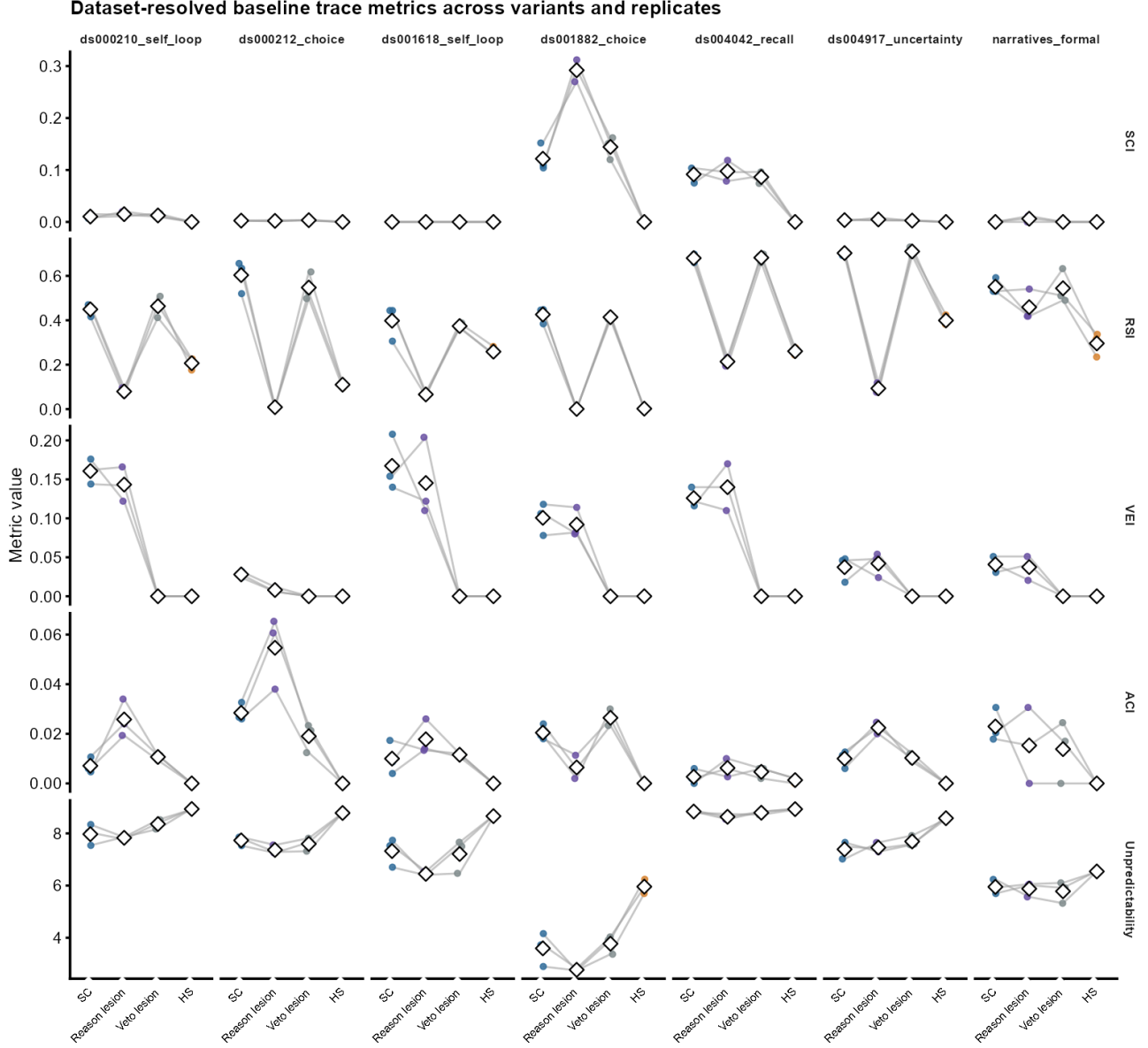


Figure 1: Dataset-resolved baseline trace metrics across variants and replicate indices. Points show replicate-level observations, connecting segments show within-replicate variant structure, and diamonds mark dataset-level means. The display expands the main-text baseline dissociation by showing SCI, RSI, VEI, ACI and action unpredictability across all dataset namespaces. SCI, RSI, VEI and ACI denote self-continuity, reason-sensitivity, veto-effect and action-continuity indices.

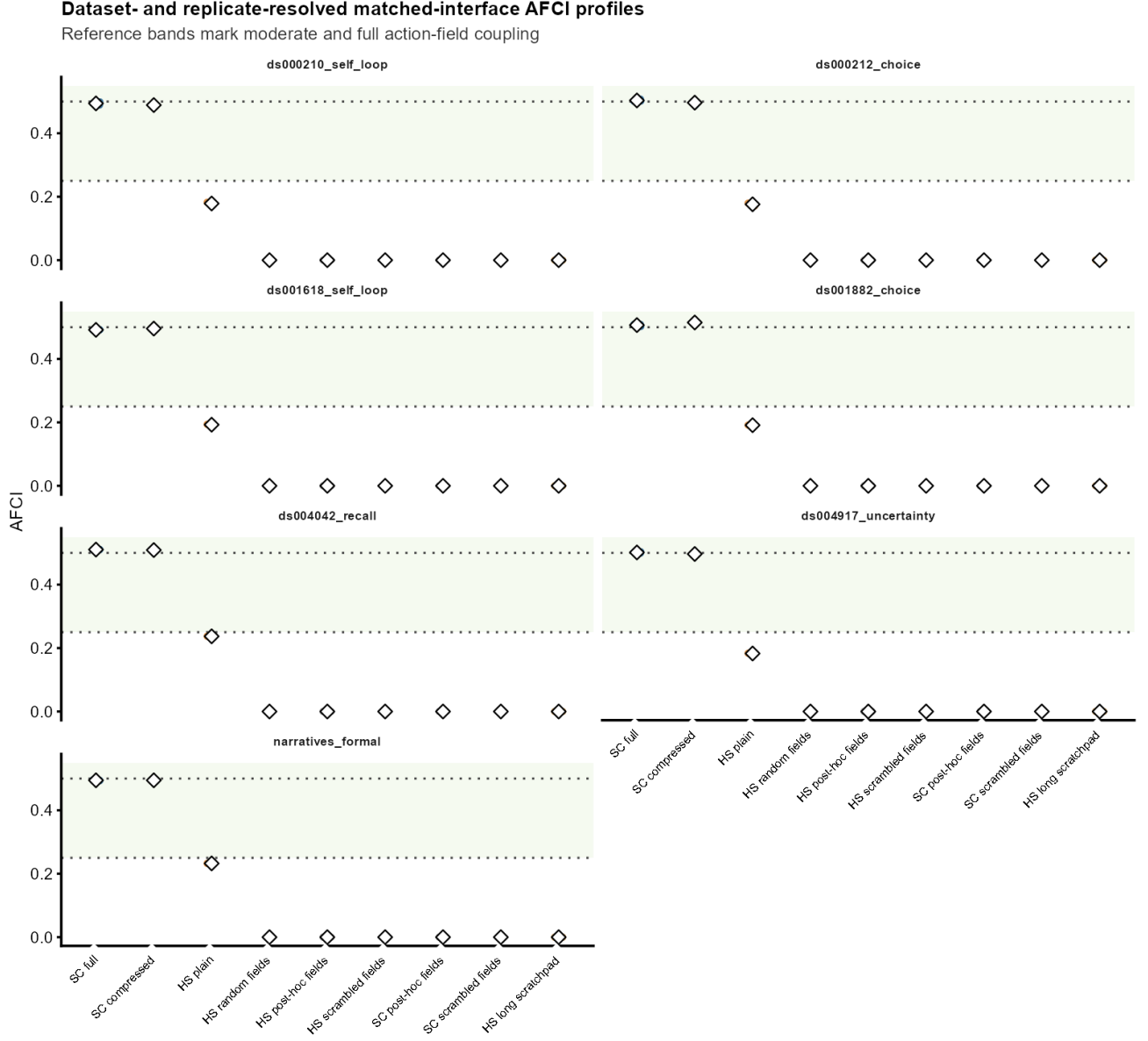


Figure 2: Dataset- and replicate-resolved matched-interface AFCI profiles. Points show replicate-level AFCI values, diamonds mark dataset means and horizontal reference bands mark moderate and full action-field coupling regions. AFCI denotes action-field coupling index. The compressed structured-control sensitivity variant is shown as a positive sensitivity variant excluded from the predefined control-comparator set; the predefined controls are the stochastic, post-hoc, scrambled and long-scratchpad matched-interface variants.

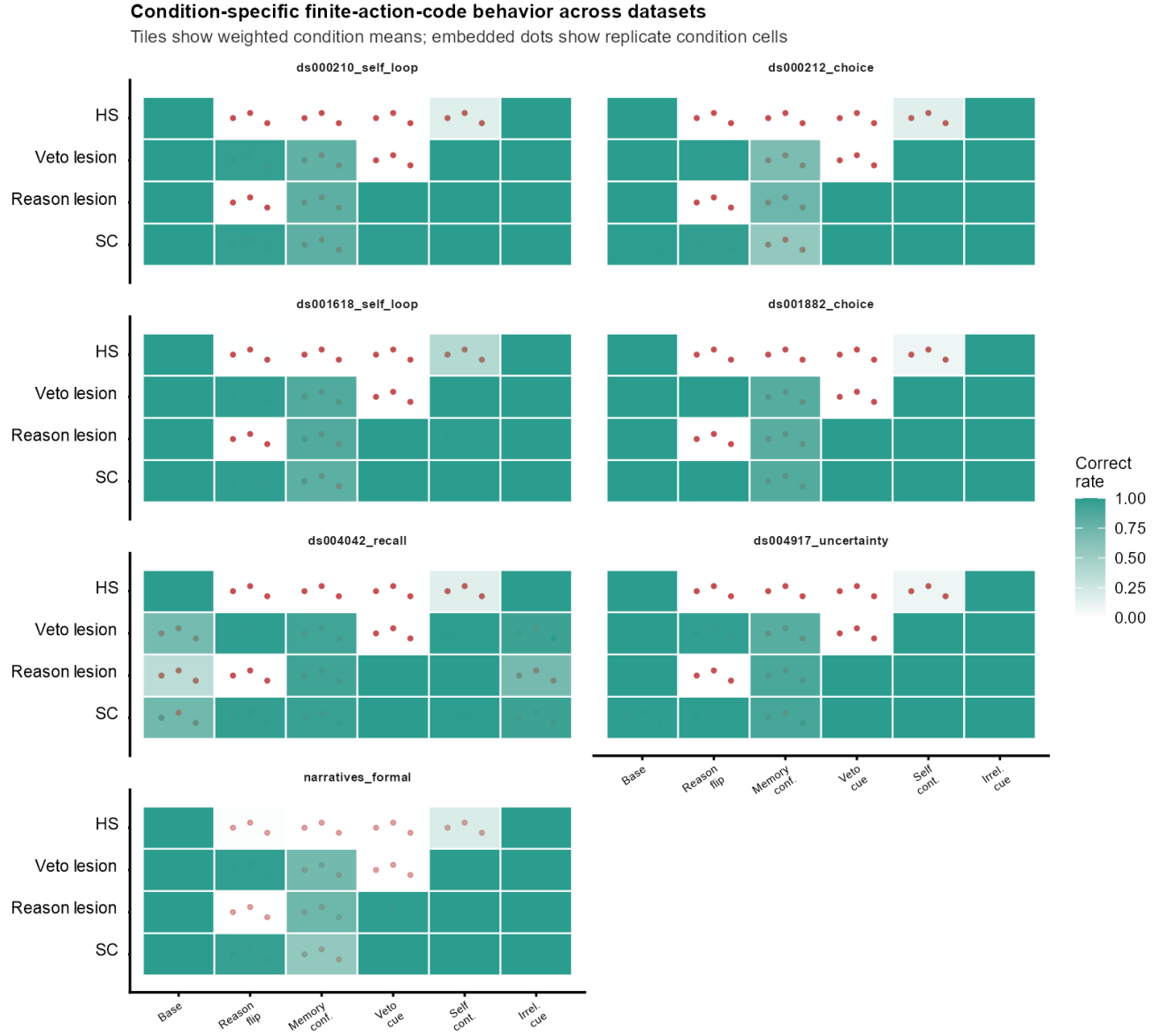


Figure 3: Condition-specific finite-action-code behavior across datasets. Tiles report weighted expected-response rates under the finite experimental action alphabet, while embedded points show replicate condition cells within each dataset. Expected labels are scorer-side annotations and are not visible to the model-visible prompt.

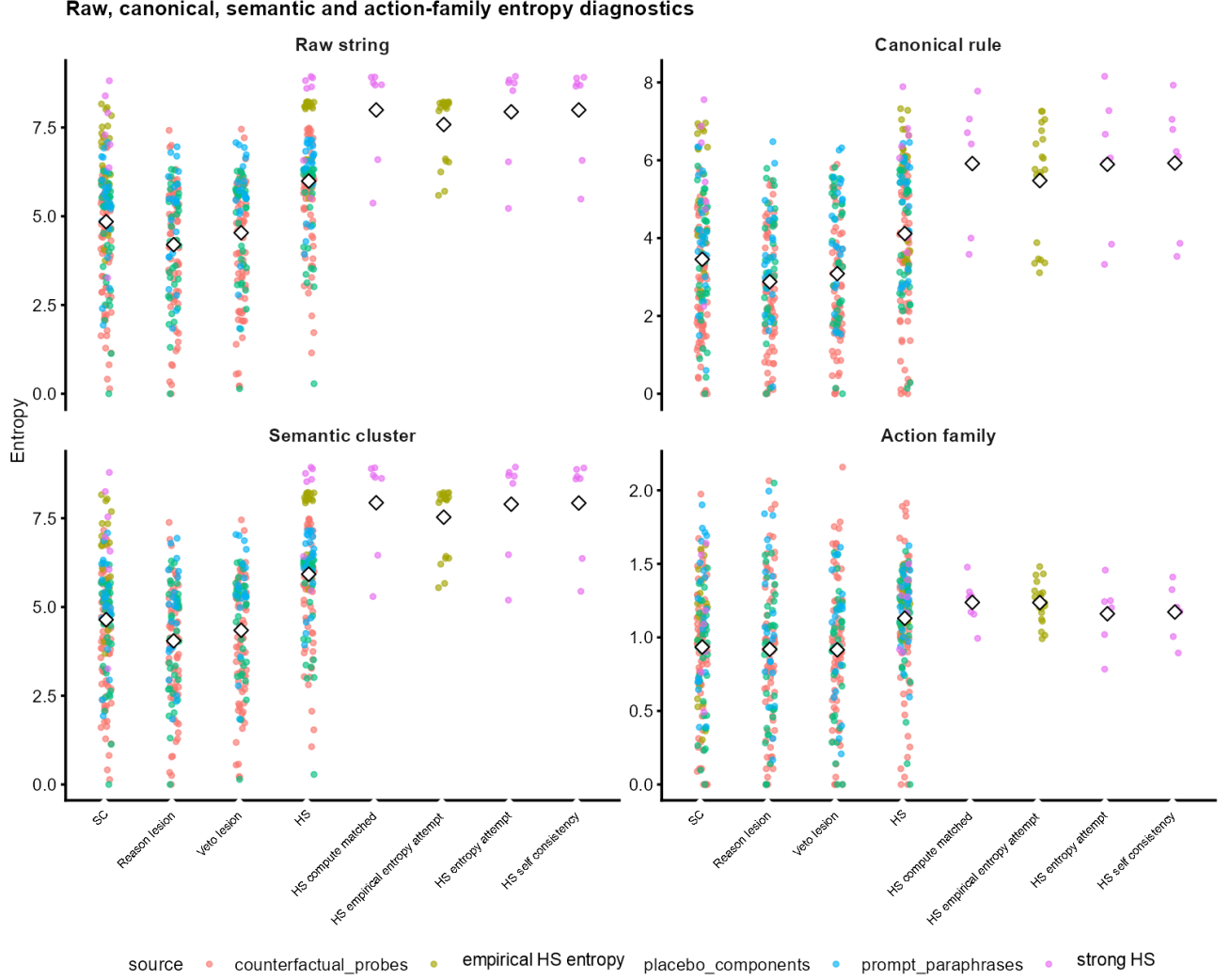


Figure 4: Raw, canonical, semantic and action-family entropy diagnostics. Points show retained entropy rows by source and variant, and diamonds mark variant means within entropy layer. The predefined 0.15 closeness criterion was not satisfied in the retained calibration attempts; these panels report calibration distance under the retained attempts.

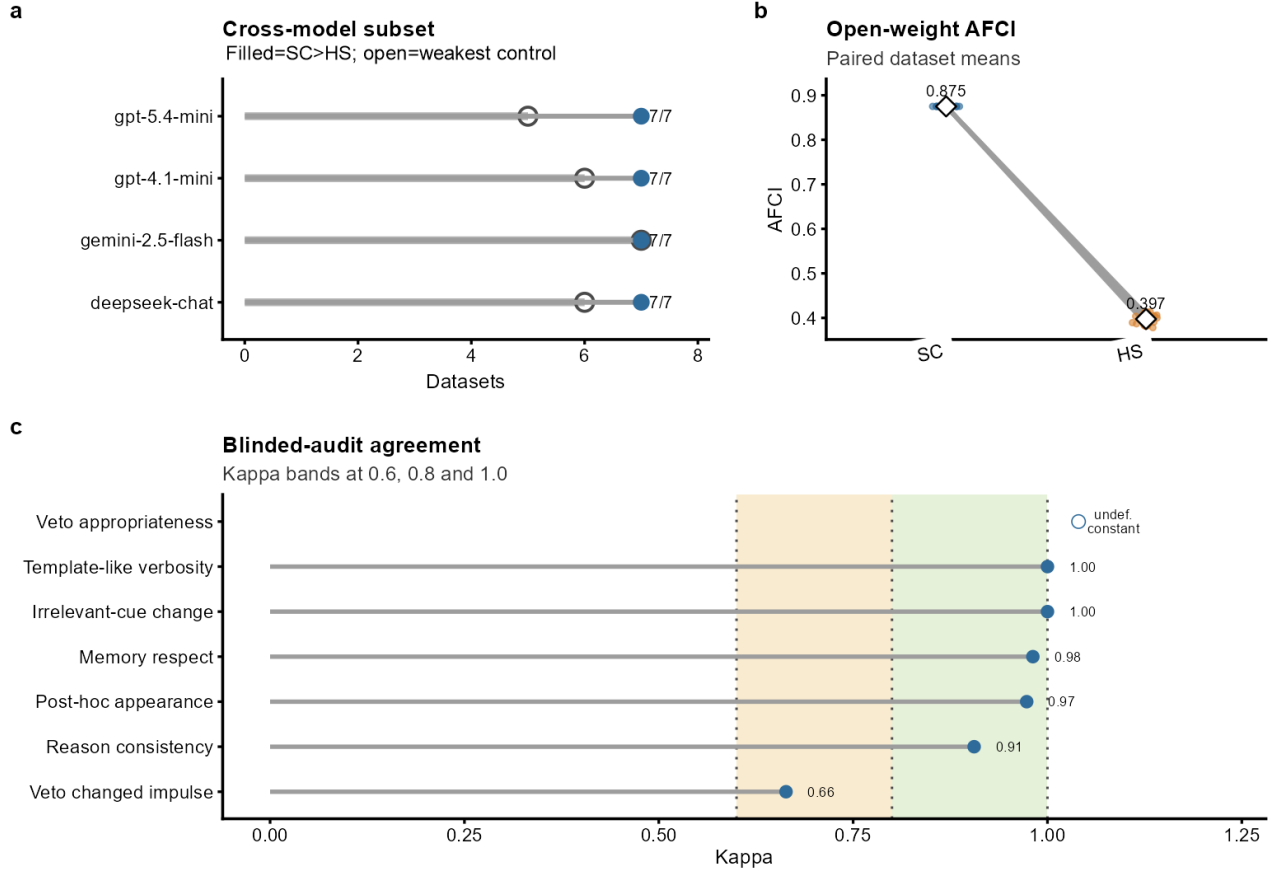


Figure 5: Auxiliary cross-model, open-weight and blinded-audit validation. The cross-model panel reports structured-control greater-than high-stochasticity dataset support while marking weaker structured-control counts separately; the open-weight panel reports paired dataset means for the Qwen2.5-14B-Instruct reference model; and the audit panel reports output-level annotation reliability display under blinding to variant identity. The open-weight validation matrix, formal API validation, strict-lesion slice, calibration and matching criteria and 1,200-item blinded audit are summarized in the main validation figure and Supplementary Table 33.

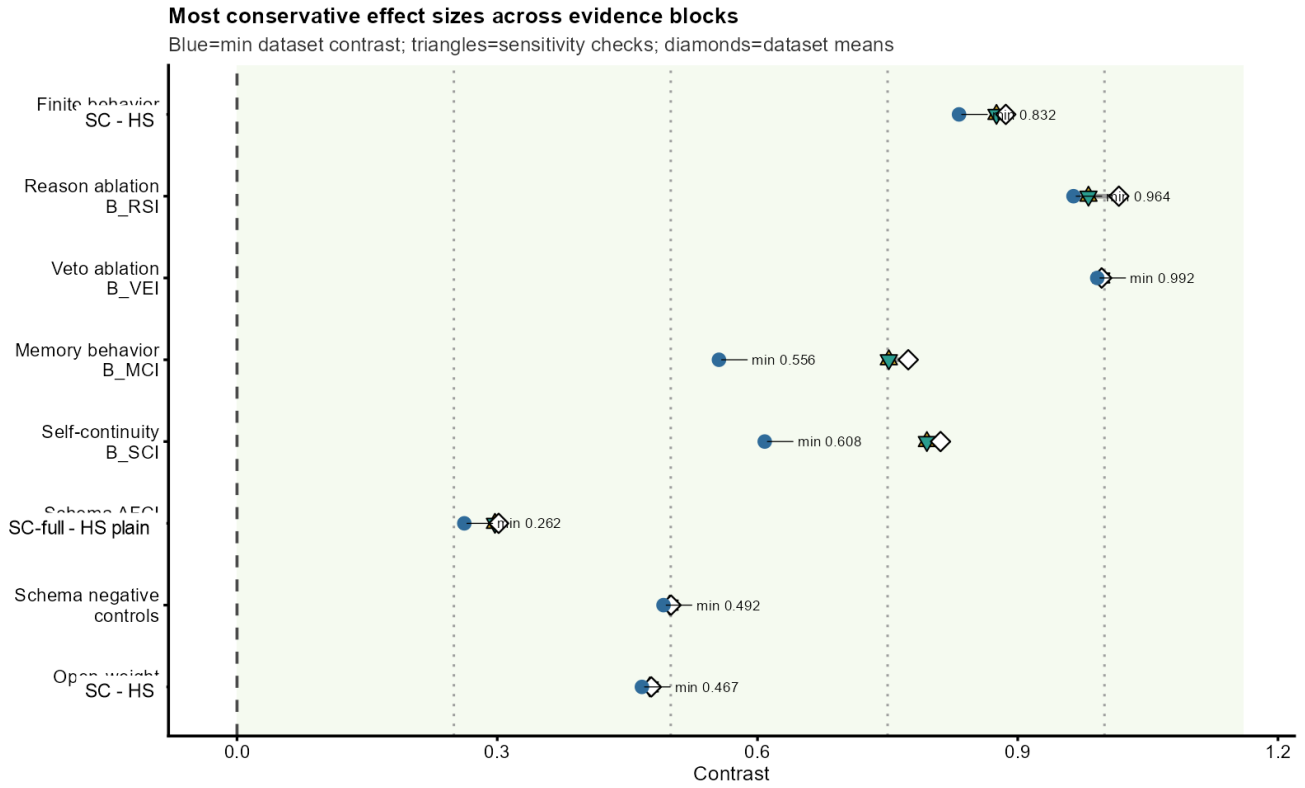


Figure 6: Most conservative effect-size summaries across evidence blocks. Blue points report minimum dataset-level contrasts, triangles report leave-one-dataset-out and top-effect-removed summaries, and diamonds report dataset means for the baseline, matched-interface, finite-action-code and open-weight evidence blocks. The display places effect-size support alongside direction counts.

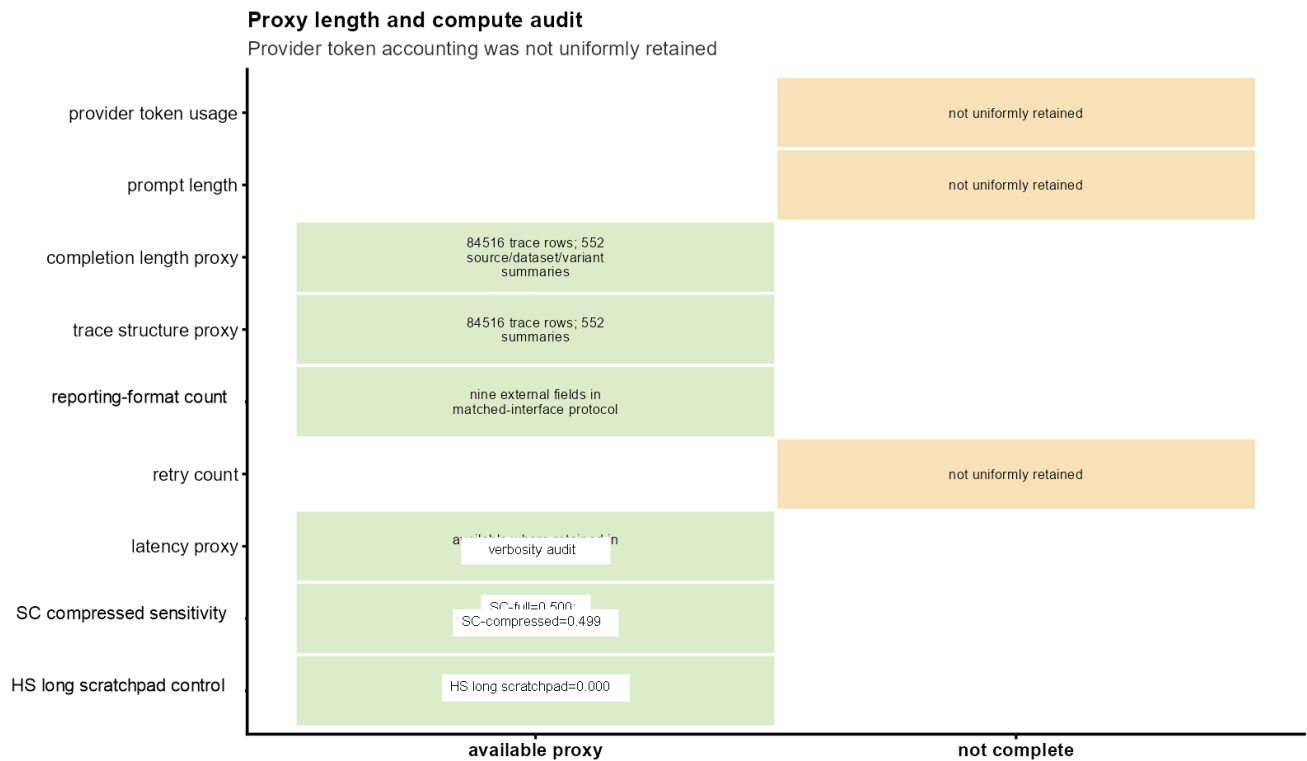


Figure 7: Proxy length and compute audit. The figure summarizes prompt-length, completion-length, field-count, retry/interface-diagnostic and compressed structured-control sensitivity records used to bound verbosity and compute explanations. Provider-side token metadata were not uniformly retained; the display provides proxy length/structure evidence without complete provider-token accounting or token-matched inference.

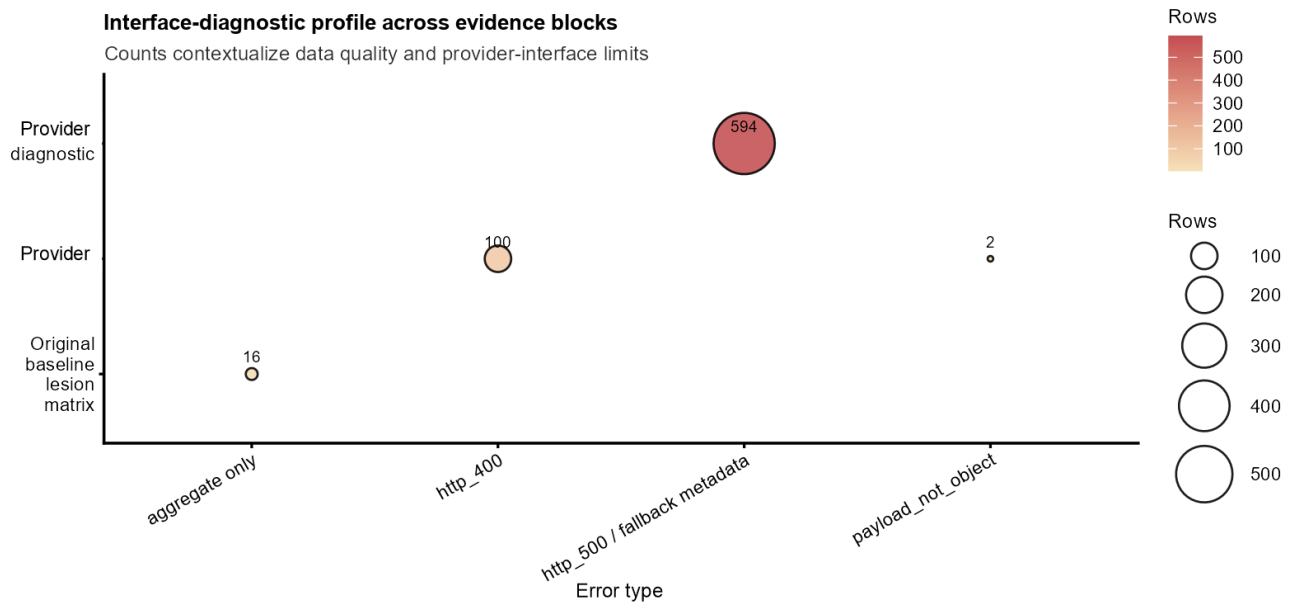


Figure 8: Error-accounting profile across evidence blocks. Bubble area and labels report error-accounting rows for final-evidence-compatible interface diagnostics, non-final provider-interface attempts and entropy-calibration diagnostics. These counts contextualize data quality and provider-interface limitations separately from the primary effect estimates.

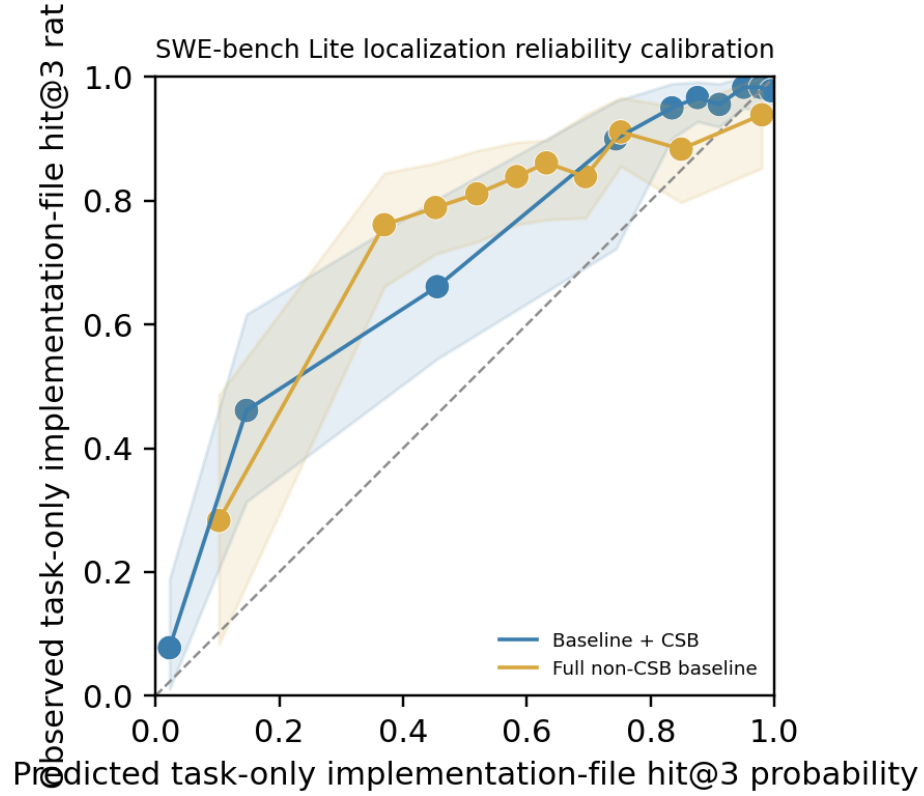


Figure 9: SWE-bench Lite issue-to-file reliability calibration. Curves bin issue-grouped five-fold predicted probabilities from the full non-CSB baseline and the baseline augmented with causal state binding. Points show decile means over 1,800 model-issue records; point size is proportional to the number of records in the bin. Shaded bands show issue-cluster bootstrap 95% intervals, and the diagonal is perfect calibration. This display calibrates the full 300-record issue-to-file localization endpoint rather than SWE-bench issue resolution.

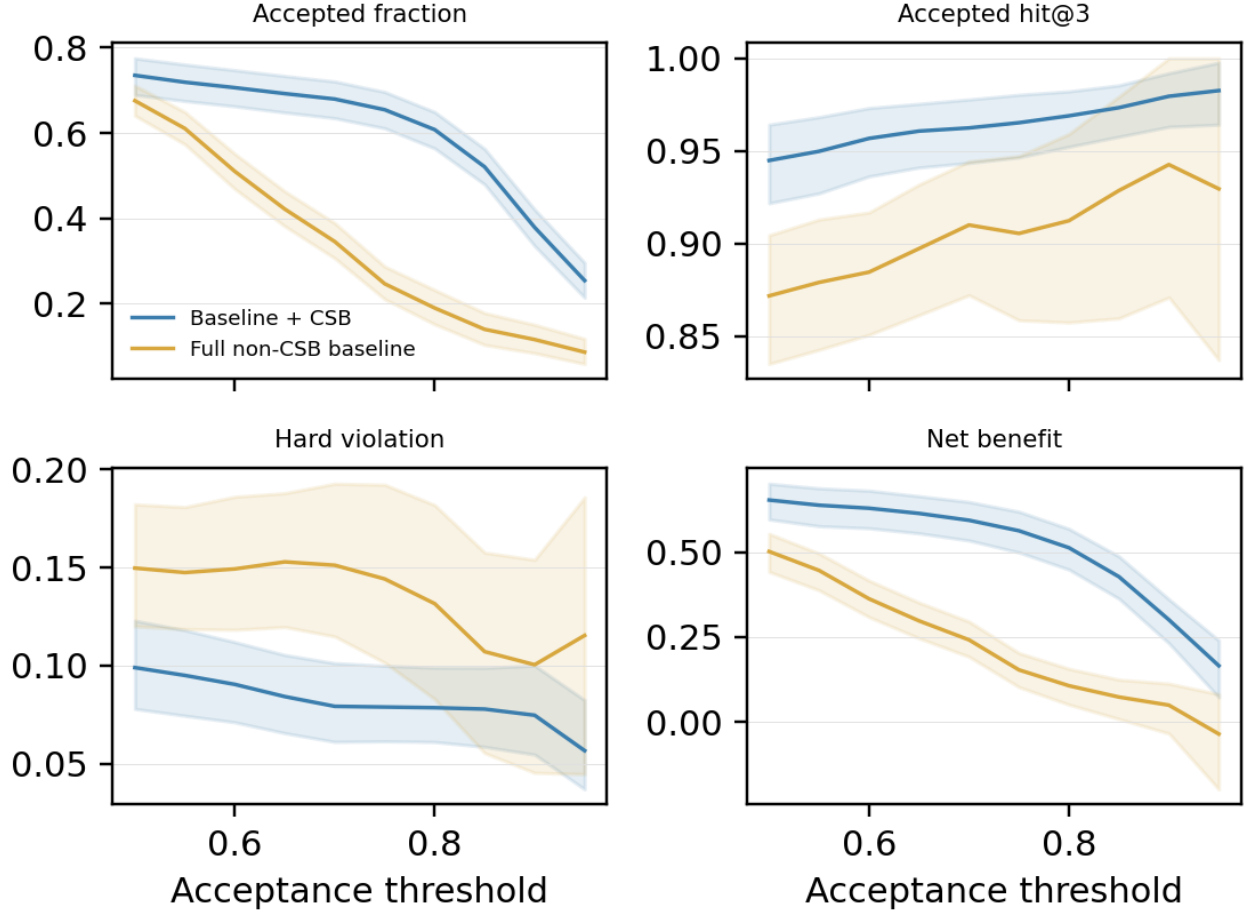


Figure 10: SWE-bench Lite thresholded localization-guard analysis. Issue-grouped five-fold predicted task-only implementation-file hit@3 probabilities were thresholded to accept or reject model-issue localization records. Panels show accepted fraction, accepted task-only hit@3, task-only hard-constraint violation rate and net benefit across thresholds for the full gold-free non-CSB baseline and the same baseline augmented with causal state binding. Shaded bands show issue-cluster bootstrap 95% intervals. The analysis tests reliability triage behavior for issue-to-file localization; patch generation and test-suite execution are outside this endpoint.

Table 1: Variant protocol-identifier registry. Canonical labels are raw protocol identifiers retained for reproducibility; alias notes identify export names or descriptive names that appear in derived result exports.

Canonical variant	Used in	Definition	Export aliases or descriptive names	Role
A4	baseline, behavior, strong controls	structured variant with branching, self-state, memory, reason graph and veto enabled	none	primary structured variant
A5	baseline, behavior, strong controls	stochastic comparator with branching and high-temperature sampling while self/memory/reason/veto are disabled	none	primary stochastic comparator
A4_no_reason	baseline, behavior, robustness	A4 with reason component removed	none	reason-ablation comparator
A4_no_veto	baseline, behavior, robustness	A4 with veto component removed	none	veto-ablation comparator
A4_full	matched-interface AFCI	full structured matched-interface variant	none	positive matched-interface variant
A4_compressed	matched-interface AFCI	compressed positive sensitivity variant excluded from the predefined control-comparator set	none	verbosity sensitivity
A5_plain	matched-interface AFCI	plain stochastic matched-interface comparator	none	predefined control comparator
A5_schema_random_fields	matched-interface AFCI	A5 matched-interface fields generated as random or template fields	random-field schema A5	predefined control comparator
A5_schema_posthoc_fields	matched-interface AFCI and open-weight reference model	A5 final action generated before post-hoc fields	post-hoc schema A5; A5_schema in open-weight runtime exports	predefined control comparator
A5_schema_scrambled_fields	matched-interface AFCI	A5 fields scrambled from another event/action source	scrambled-field schema A5	predefined control comparator
A4_posthoc_fields	matched-interface AFCI	A4 final action retained while fields are generated after final-action selection	A4 post-hoc fields	predefined control comparator
A4_scrambled_fields	matched-interface AFCI	A4 final action retained while fields are paired with wrong event/action information	A4 scrambled fields	predefined control comparator
A5_long_scratchpad	matched-interface AFCI	A5 stochastic comparator with extended scratchpad/field budget	long-scratchpad A5	predefined control comparator
A5_entropy_attempt	strong stochastic controls, entropy diagnostics	lower-temperature stochastic attempt	lower-temperature stochastic-attempt export alias	predefined entropy-closeness criterion unsatisfied
A5_empirical_entropy_attempt	entropy-calibration diagnostic	selected temperature/top-p candidate from empirical calibration diagnostic	calibration candidate aliases A5_attempt_t02_p07 A5_attempt_t020_p075	predefined entropy-closeness criterion unsatisfied

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Table 1: Variant protocol-identifier registry. Canonical labels are raw protocol identifiers retained for reproducibility; alias notes identify export names or descriptive names that appear in derived result exports. (continued)

Canonical variant	Used in	Definition	Export aliases or descriptive names	Role
A5_compute matched	strong stochastic controls	compute-control attempt under the implemented protocol; raw identifier does not indicate strict token/compute matching	none	strong stochastic comparator
A5_self consistency	strong stochastic controls	self-consistency stochastic control under the implemented protocol	none	strong stochastic comparator

Table 2: Mechanistic claim components, operational tests and contrary results.

Table role	Claim component	Operational test	Contrary result	Interpretive boundary
mechanism_map	stochastic contrast	The high-stochasticity comparator increases action unpredictability while disabling self, memory, reason and veto modules.	A strong stochastic control recovers structured-control metrics.	Operational stochastic-substitution test; no claim about consciousness, free will or neural homology.
mechanism_map	matched-interface action-field coupling	All controls use the same reason/memory/self/veto reporting format and are scored by AFCl.	Post-hoc, scrambled, random-field or verbosity controls match the structured matched-interface AFCl.	Tests reporting-format availability under a shared interface.
mechanism_map	finite-action-code behavior	Scorer-side records define hidden expected labels and score only canonical final action codes.	The structured-control variant does not pass behavioral criteria or ablations preserve the full structured-control profile.	Fixed action ontology; does not test unconstrained natural-language decision making.

S1. Implemented Agent Family and Prompt/Event Corpora

The implemented family is defined by explicit module toggles and sampling settings. The table reports the operational variant definitions used by the lesion tests.

Table 3: Implemented agent-family variants and operational roles.

Variant	Branch	Self-state	Memory	Reason graph	Veto	Temperature	Role
A0	no	no	no	no	no	0.1	minimal non-structured reference
A1	yes	no	no	no	no	0.2	branching-only reference
A2	yes	yes	no	yes	no	0.2	self plus reason without memory/veto
A3	yes	yes	yes	yes	no	0.2	memory-enabled structured variant without veto
A4	yes	yes	yes	yes	yes	0.2	primary structured variant
A4_no_reason	yes	yes	yes	no	yes	0.2	reason-graph lesion
A4_no_veto	yes	yes	yes	yes	no	0.2	veto lesion
A5	yes	no	no	no	no	0.9	primary stochastic comparator
A5_entropy attempt	yes	no	no	no	no	0.2	lower-temperature stochastic attempt
compute-control attempt	yes	no	no	no	no	0.9	compute-control attempt; raw identifier retained in Table 1
A5_self consistency	yes	no	no	no	no	0.9	multi-sample stochastic control
A5_empirical entropy_attempt	yes	no	no	no	no	1	empirical entropy-attempt comparator

Table 4: Dataset namespaces used as prompt/event corpora.

Dataset namespace	Prompt-corpus role	Construct pressure
narratives_formal	narrative prompt corpus	agency and self-narrative cues
ds000210_self_loop	self/prospection prompt corpus	self-continuity
ds001618_self_loop	autobiographical/self prompt corpus	self-memory loop
ds001882_choice	choice prompt corpus	reason/value conflict
ds000212_choice	choice prompt corpus	reason, veto and agency
ds004917_uncertainty	uncertainty/risk prompt corpus	stochastic contrast
ds004042_recall	recall/memory prompt corpus	memory constraint

S2. Baseline Lesion Matrix

The baseline lesion matrix evaluates whether output dispersion and structured-control traces separate under the implemented structured-control/high-stochasticity lesion design. Raw protocol identifiers are retained in the tables for exact export matching. The criterion summary records the realized call count and dataset-direction outcomes.

Table 5: Baseline lesion-matrix criterion summary.

Criterion	Field	Value
baseline lesion matrix	status	pass
baseline lesion matrix	profile	full
baseline lesion matrix	calls	74352
baseline lesion matrix	remote calls	74336
baseline lesion matrix	unrecovered generations	16
baseline lesion matrix	unrecovered-generation rate	0.000215193
baseline lesion matrix	model	gpt-5.4-mini
baseline lesion matrix	fallback model	qwen3-30b-a3b
baseline lesion matrix	structured-control SCI > high-stochasticity datasets	5
baseline lesion matrix	high-stochasticity unpredictability > structured-control datasets	7
baseline lesion matrix	reason-lesion drop datasets	7
baseline lesion matrix	veto-lesion drop datasets	7

Table 6: Baseline trace metrics summarized by variant.

Variant	SCI mean	RSI mean	VEI mean	ACI mean	Unpredictability mean	Distance to structured mean	Rows
A0	0	0.14931	0	9.524e-05	8.073	1.59321	21
A1	0	0.18819	0	0	8.00038	0.476597	21
A2	0.00423593	0.788509	9.524e-05	0	7.99426	0.320232	21
A3	0.0378864	0.537366	0	0.00956706	7.15695	0.172573	21
A4	0.0328108	0.544336	0.0944023	0.014518	6.9708	0	21
A4_no_reason	0.0596636	0.131598	0.0868688	0.0212342	6.62649	0.474022	21
A4_no_veto	0.0355957	0.533555	0	0.0137792	7.0287	0.181687	21
A5	0	0.219036	0	0.000190476	8.05561	0.447575	21

Table 7: Baseline trace metrics by dataset namespace, variant and replicate index.

Dataset	Replicate index	Variant	SCI	RSI	VEI	ACI	Unpredictability	Distance to structured
ds000210_self_loop	1	A0	0	0.328	0	0	8.76825	1.35526
ds000210_self_loop	1	A1	0	0.144	0	0	8.89627	0.442842
ds000210_self_loop	1	A2	0.001	0.648	0	0	8.94578	0.245967
ds000210_self_loop	1	A3	0.0144	0.462	0	0.0106667	8.16901	0.162433
ds000210_self_loop	1	A4	0.007	0.47	0.162	0.006	8.03157	0
ds000210_self_loop	1	A4_no_reason	0.02	0.096	0.166	0.0193333	7.77433	0.393241
ds000210_self_loop	1	A4_no_veto	0.0114	0.412	0	0.00933333	8.38953	0.212166
ds000210_self_loop	1	A5	0	0.218	0	0	8.93627	0.315964
ds000210_self_loop	2	A0	0	0.344	0	0	8.76751	1.21486
ds000210_self_loop	2	A1	0	0.14	0	0	8.87448	0.346922
ds000210_self_loop	2	A2	0.001	0.66	0	0	8.94178	0.356704
ds000210_self_loop	2	A3	0.014	0.466	0	0.006	8.51228	0.249926
ds000210_self_loop	2	A4	0.015	0.416	0.144	0.0106667	7.53976	0
ds000210_self_loop	2	A4_no_reason	0.014	0.064	0.142	0.024	7.86176	0.367291
ds000210_self_loop	2	A4_no_veto	0.01	0.47	0	0.0113333	8.16759	0.213498
ds000210_self_loop	2	A5	0	0.226	0	0	8.95378	0.292231
ds000210_self_loop	3	A0	0	0.36	0	0	8.83731	1.32969
ds000210_self_loop	3	A1	0	0.19	0	0	8.88506	0.397445
ds000210_self_loop	3	A2	0.001	0.688	0	0	8.91476	0.330203
ds000210_self_loop	3	A3	0.013	0.468	0	0.0113333	8.46181	0.179912
ds000210_self_loop	3	A4	0.009	0.462	0.176	0.00466667	8.33933	0
ds000210_self_loop	3	A4_no_reason	0.011	0.078	0.122	0.034	7.85439	0.408794
ds000210_self_loop	3	A4_no_veto	0.016	0.508	0	0.0113333	8.52001	0.18609
ds000210_self_loop	3	A5	0	0.176	0	0	8.91476	0.35543
ds000212_choice	1	A0	0	0.296	0	0	8.72665	1.15236
ds000212_choice	1	A1	0	0.102	0	0	8.7556	0.657512
ds000212_choice	1	A2	0	0.876	0	0	8.71947	0.242358
ds000212_choice	1	A3	0.0054	0.434	0	0.011	7.07074	0.245605
ds000212_choice	1	A4	0.00252	0.656	0.028	0.0266667	7.85523	0
ds000212_choice	1	A4_no_reason	0.00108	0.012	0.006	0.0606667	7.55452	0.680466
ds000212_choice	1	A4_no_veto	0.00396	0.526	0	0.0233333	7.81526	0.146769
ds000212_choice	1	A5	0	0.116	0	0	8.77971	0.541392
ds000212_choice	2	A0	0	0.246	0	0	8.7878	1.18602
ds000212_choice	2	A1	0	0.086	0	0	8.69956	0.724426
ds000212_choice	2	A2	0.00036	0.884	0	0	8.76596	0.313144
ds000212_choice	2	A3	0.00216	0.492	0	0.018	7.55397	0.172383
ds000212_choice	2	A4	0.00216	0.634	0.032	0.026	7.80838	0
ds000212_choice	2	A4_no_reason	0.0036	0.01	0.012	0.038	7.2857	0.671147
ds000212_choice	2	A4_no_veto	0.00252	0.498	0	0.0124	7.31826	0.166749
ds000212_choice	2	A5	0	0.098	0	0	8.79493	0.53777
ds000212_choice	3	A0	0	0.296	0	0	8.83555	1.0744
ds000212_choice	3	A1	0	0.09	0	0	8.6902	0.591411
ds000212_choice	3	A2	0	0.868	0	0	8.75269	0.397617
ds000212_choice	3	A3	0.00144	0.63	0	0.0253333	7.79394	0.119062
ds000212_choice	3	A4	0.00288	0.52	0.024	0.0326667	7.53217	0
ds000212_choice	3	A4_no_reason	0.00144	0.004	0.006	0.0653333	7.25292	0.541525
ds000212_choice	3	A4_no_veto	0.0036	0.618	0	0.0213333	7.65724	0.168122
ds000212_choice	3	A5	0	0.116	0	0	8.77164	0.406437
ds001618_self_loop	1	A0	0	0.112	0	0	8.47286	1.35064
ds001618_self_loop	1	A1	0	0.19	0	0	8.61063	0.428248
ds001618_self_loop	1	A2	0	0.72	0.002	0	8.77191	0.320151
ds001618_self_loop	1	A3	0	0.362	0	0.005	7.81575	0.274838
ds001618_self_loop	1	A4	0	0.444	0.154	0.0173333	7.52728	0
ds001618_self_loop	1	A4_no_reason	0	0.064	0.204	0.0133333	6.53898	0.383927
ds001618_self_loop	1	A4_no_veto	0	0.366	0	0.012	7.66101	0.317151
ds001618_self_loop	1	A5	0	0.246	0	0	8.67665	0.271338

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Table 7: Baseline trace metrics by dataset namespace, variant and replicate index. (continued)

Dataset	Replicate index	Variant	SCI	RSI	VEI	ACI	Unpredictability	Distance to structured
ds001618_self_loop	2	A0	0	0.144	0	0	8.52178	1.36285
ds001618_self_loop	2	A1	0	0.2	0	0	8.56596	0.487458
ds001618_self_loop	2	A2	0	0.694	0	0	8.77412	0.33743
ds001618_self_loop	2	A3	0	0.368	0	0.0085	7.72458	0.17737
ds001618_self_loop	2	A4	0	0.444	0.14	0.00866667	7.73685	0
ds001618_self_loop	2	A4_no_reason	0	0.076	0.122	0.026	6.41047	0.402855
ds001618_self_loop	2	A4_no_veto	0	0.368	0	0.012	6.46868	0.166994
ds001618_self_loop	2	A5	0	0.248	0	0	8.63753	0.298441
ds001618_self_loop	3	A0	0	0.106	0	0	8.43722	1.43727
ds001618_self_loop	3	A1	0	0.184	0	0	8.45131	0.580472
ds001618_self_loop	3	A2	0	0.596	0	0	8.76532	0.413091
ds001618_self_loop	3	A3	0	0.396	0	0	7.68315	0.226955
ds001618_self_loop	3	A4	0	0.306	0.208	0.00333333	6.70393	0
ds001618_self_loop	3	A4_no_reason	0	0.058	0.11	0.014	6.39787	0.334682
ds001618_self_loop	3	A4_no_veto	0	0.388	0	0.0106667	7.50181	0.226884
ds001618_self_loop	3	A5	0	0.282	0	0	8.66935	0.263545
ds001618_self_loop	3	A0	0	0.004	0	0	6.52926	1.94798
ds001882_choice	1	A0	0	0.004	0	0	5.99445	0.479681
ds001882_choice	1	A1	0	0.004	0	0	5.5912	0.361726
ds001882_choice	1	A2	0	0.75	0	0	3.81626	0.118239
ds001882_choice	1	A3	0.008	0.396	0	0.008	3.72608	0
ds001882_choice	1	A4	0.162	0.446	0.106	0.0193333	2.82488	0.462926
ds001882_choice	1	A4_no_reason	0.27	0	0.08	0.002	2.82488	0.113551
ds001882_choice	1	A4_no_veto	0.12	0.424	0	0.03	4.03051	0.113551
ds001882_choice	1	A5	0	0	0	0	5.9395	0.998752
ds001882_choice	2	A0	0	0.002	0	0	6.40637	1.95812
ds001882_choice	2	A1	0	0.002	0	0	5.64805	0.476378
ds001882_choice	2	A2	0.007	0.768	0	0	5.77716	0.363858
ds001882_choice	2	A3	0.141	0.468	0	0.015	4.70427	0.125849
ds001882_choice	2	A4	0.104	0.448	0.118	0.024	4.15845	0
ds001882_choice	2	A4_no_reason	0.312	0.002	0.114	0.006	2.74187	0.492869
ds001882_choice	2	A4_no_veto	0.151	0.404	0	0.0233333	3.91573	0.13466
ds001882_choice	2	A5	0	0.004	0	0	6.24003	0.696129
ds001882_choice	3	A0	0	0	0	0	6.52098	1.92553
ds001882_choice	3	A1	0	0.006	0	0	5.86191	0.406223
ds001882_choice	3	A2	0.02	0.756	0	0	5.22555	0.402404
ds001882_choice	3	A3	0.145	0.45	0	0.012	3.74179	0.108793
ds001882_choice	3	A4	0.109	0.384	0.078	0.018	2.89757	0
ds001882_choice	3	A4_no_reason	0.294	0	0.082	0.0113333	2.74725	0.426429
ds001882_choice	3	A4_no_veto	0.162	0.414	0	0.026	3.36946	0.0994636
ds001882_choice	3	A5	0	0.002	0	0	5.69531	0.945267
ds004042_recall	1	A0	0	0.03	0	0	8.3718	0.14564
ds004042_recall	1	A1	0	0.192	0	0	8.39222	0
ds004042_recall	1	A2	0.012952	0.902	0	0	8.54148	0.496309
ds004042_recall	1	A3	0.100937	0.65	0.14	0.002	8.83922	0
ds004042_recall	1	A4	0.103603	0.682	0.14	0.006	8.54148	0.16193
ds004042_recall	1	A4_no_reason	0.0956766	0.194	0	0.002	8.82551	0.471786
ds004042_recall	1	A4_no_veto	0.0962956	0.66	0	0	8.94578	0.227454
ds004042_recall	1	A5	0	0.244	0	0	8.9057	0.551628
ds004042_recall	2	A0	0	0.026	0	0.002	8.95378	0.277931
ds004042_recall	2	A1	0	0.218	0	0	8.3718	0.14564
ds004042_recall	2	A2	0.0092848	0.87	0	0	8.39222	0
ds004042_recall	2	A3	0.091722	0.72	0	0	8.54148	0.496309
ds004042_recall	2	A4	0.0748846	0.698	0.116	0.006	8.82551	0.16193
ds004042_recall	2	A4_no_reason	0.118549	0.232	0.17	0.00266667	8.94578	0.227454
ds004042_recall	2	A4_no_veto	0.0741564	0.688	0	0.006	8.9057	0.551628
ds004042_recall	2	A5	0	0.244	0	0	8.95378	0.277931
ds004042_recall	3	A0	0	0.026	0	0	8.3718	0.14564
ds004042_recall	3	A1	0	0.218	0	0	8.39222	0
ds004042_recall	3	A2	0.0092848	0.87	0	0	8.54148	0.496309
ds004042_recall	3	A3	0.091722	0.72	0	0	8.82551	0.16193
ds004042_recall	3	A4	0.0748846	0.698	0.116	0.006	8.94578	0.227454
ds004042_recall	3	A4_no_reason	0.118549	0.232	0.17	0.00266667	8.9057	0.551628
ds004042_recall	3	A4_no_veto	0.0741564	0.688	0	0.006	8.95378	0.277931
ds004042_recall	3	A5	0	0.244	0	0	8.3718	0.14564

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Table 7: Baseline trace metrics by dataset namespace, variant and replicate index. (continued)

Dataset	Replicate index	Variant	SCI	RSI	VEI	ACI	Unpredictability	Distance to structured
ds004042_recall	2	A5	0	0.276	0	0.002	8.91306	0.446837
ds004042_recall	3	A0	0	0.032	0	0	8.91048	1.941
ds004042_recall	3	A1	0	0.226	0	0	8.94027	0.492784
ds004042_recall	3	A2	0	0.904	0	0	8.94427	0.287693
ds004042_recall	3	A3	0.0276378	0.724	0	0.01	8.82833	0.144852
ds004042_recall	3	A4	0.0965382	0.66	0.122	0	8.87975	0
ds004042_recall	3	A4_no_reason	0.0786418	0.216	0.11	0.01	8.62266	0.458524
ds004042_recall	3	A4_no_veto	0.0886568	0.698	0	0.006	8.83318	0.133679
ds004042_recall	3	A5	0	0.262	0	0.002	8.92978	0.428004
ds004917_uncertainty	1	A0	0	0.018	0	0	8.62299	2.0015
ds004917_uncertainty	1	A1	0	0.338	0	0	8.63624	0.400878
ds004917_uncertainty	1	A2	0	0.912	0	0	8.43571	0.494227
ds004917_uncertainty	1	A3	0.00036	0.682	0	0.013	8.09108	0.161046
ds004917_uncertainty	1	A4	0.00324	0.692	0.046	0.0113333	7.01934	0
ds004917_uncertainty	1	A4_no_reason	0.00432	0.076	0.048	0.0246667	7.65113	0.616474
ds004917_uncertainty	1	A4_no_veto	0.00288	0.692	0	0.0088	7.91809	0.0461257
ds004917_uncertainty	1	A5	0.00324	0.382	0	0	8.54235	0.455262
ds004917_uncertainty	2	A0	0	0.012	0	0	8.51087	2.13843
ds004917_uncertainty	2	A1	0	0.336	0	0	8.40745	0.387633
ds004917_uncertainty	2	A2	0	0.912	0	0	8.47606	0.382314
ds004917_uncertainty	2	A3	0.00036	0.69	0	0.006	7.58568	0.0960863
ds004917_uncertainty	2	A4	0.00468	0.712	0.048	0.0126667	7.65424	0
ds004917_uncertainty	2	A4_no_reason	0.00432	0.086	0.024	0.0226667	7.29729	0.667423
ds004917_uncertainty	2	A4_no_veto	0.00324	0.73	0	0.012	7.56667	0.117761
ds004917_uncertainty	2	A5	0.00252	0.392	0	0	8.65413	0.391514
ds004917_uncertainty	3	A0	0	0.004	0	0	8.60747	2.05311
ds004917_uncertainty	3	A1	0	0.304	0	0	8.61613	0.414245
ds004917_uncertainty	3	A2	0	0.916	0	0	8.47654	0.331396
ds004917_uncertainty	3	A3	0.00324	0.692	0	0.0173333	7.85269	0.0245462
ds004917_uncertainty	3	A4	0.0018	0.704	0.018	0.006	7.49391	0
ds004917_uncertainty	3	A4_no_reason	0.00828	0.118	0.054	0.02	7.4201	0.64978
ds004917_uncertainty	3	A4_no_veto	0.00216	0.708	0	0.01	7.58817	0.151182
ds004917_uncertainty	3	A5	0	0.424	0	0	8.54861	0.447065
narratives_formal	1	A0	0	0.265306	0	0	6.49726	1.28397
narratives_formal	1	A1	0	0.346939	0	0	6.53308	0.387361
narratives_formal	1	A2	0	0.72449	0	0	6.5943	0.221234
narratives_formal	1	A3	0.00742347	0.530612	0	0.0102041	6.02814	0.220534
narratives_formal	1	A4	0	0.530612	0.0510204	0.0178571	5.56579	0
narratives_formal	1	A4_no_reason	0	0.418367	0.0510204	0.0153061	5.32241	0.317218
narratives_formal	1	A4_no_veto	0	0.632653	0	0.024898	5.2241	0.247869
narratives_formal	2	A5	0	0.316327	0	0	6.46415	1.22598
narratives_formal	2	A0	0	0.214286	0	0	6.48456	0.349778
narratives_formal	2	A1	0	0.387755	0	0	6.5943	0.260554
narratives_formal	2	A2	0	0.785714	0	0	5.92067	0.167048
narratives_formal	2	A3	0	0.581633	0	0	5.91899	0
narratives_formal	2	A4	0	0.530612	0.0306122	0.0306122	6.05625	0.137085
narratives_formal	2	A4_no_reason	0.00707041	0.540816	0.0408163	0	6.10536	0.269589
narratives_formal	2	A4_no_veto	0	0.510204	0	0	6.52537	0.218853
narratives_formal	2	A5	0	0.336735	0	0	6.48456	1.63981
narratives_formal	3	A0	0	0.295918	0	0	6.48456	0.443026
narratives_formal	3	A1	0	0.265306	0	0	6.53308	0.157421
narratives_formal	3	A2	0	0.72449	0	0	5.68077	0.371015
narratives_formal	3	A3	0.00117857	0.622449	0	0.0102041	5.69492	0
narratives_formal	3	A4	0	0.591837	0.0408163	0.0204082	6.01247	0.549153
narratives_formal	3	A4_no_reason	0.0114776	0.418367	0.0204082	0.0306122		

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Table 7: Baseline trace metrics by dataset namespace, variant and replicate index. (continued)

Dataset	Replicate index	Variant	SCI	RSI	VEI	ACI	Unpredictability	Distance to structured
narratives_formal	3	A4_no_veto	0	0.489796	0	0.0170068	5.91339	0.344546
narratives_formal	3	A5	0	0.234694	0	0	6.55349	0.369185

S3. Matched-Interface Action-Field Coupling Controls

Matched-interface controls required comparator variants to use the same reporting format and then scored action-field coupling. AFCI is an interface-level analysis that uses reported content, final action and predefined field-source information; the finite-action-code benchmark supplies the reporting-format-independent behavioral test. No metadata-free content-only AFCI sensitivity is included. The compressed structured-control sensitivity variant (`A4_compressed`) is reported as a positive sensitivity variant excluded from the seven predefined control comparators: `A5_plain`, `A5_schema_random_fields`, `A5_schema_posthoc_fields`, `A5_schema_scrambled_fields`, `A4_posthoc_fields`, `A4_scrambled_fields` and `A5_long_scratchpad`. The self-action AFCI term is sparse in the matched-interface trace scorer. The finite-action-code B_{SCI} score is defined separately over final action-code stability under the predefined self-continuity probes.

Table 8: Matched-interface control criterion summary.

Criterion	Field	Value
matched-interface controls	status	pass
matched-interface controls	dataset_count	7
matched-interface controls	control_count	7
matched-interface controls	min_a4_afci_gt_control_dataset_count	7
matched-interface controls	cluster_bootstrap_afci_delta_low95	0.451911
matched-interface controls	call_count	26946
matched-interface controls	remote calls	26946
matched-interface controls	unrecovered_count	0
matched-interface controls	unrecovered_rate	0
matched-interface controls	planned calls	26946
matched-interface controls	model	gpt-5.4-mini

Table 9: Predefined control-comparator recovery status for matched-interface AFCI.

Comparator variant	Datasets	Structured AFCI > comparator datasets	Comparator recovered structured datasets
<code>A5_plain</code>	7	7	0
<code>A5_schema_random_fields</code>	7	7	0
<code>A5_schema_posthoc_fields</code>	7	7	0
<code>A5_schema_scrambled_fields</code>	7	7	0
<code>A4_posthoc_fields</code>	7	7	0
<code>A4_scrambled_fields</code>	7	7	0
<code>A5_long_scratchpad</code>	7	7	0

Table 10: AFCI component means by matched-interface variant. *A4_compressed* is a compressed positive sensitivity variant excluded from the seven predefined control comparators.

Variant	Reason alignment mean	Memory alignment mean	Veto alignment mean	Self alignment mean	Irrelevant suppression mean	AFCI mean	AFCI + suppression mean	Rows
A4_compressed	0.491286	0.502222	1	0.00412698	1	0.499409	0.599527	21
A4_full	0.494461	0.503968	1	0.00285714	1	0.500322	0.600257	21
A4_posthoc fields	0	0	0	0	0	0	0	21
A4_scrambled fields	0	0	0	0	0	0	0	21
A5_long scratchpad	0	0	0	0	0	0	0	21
A5_plain	0.069306	0.225474	0.5	0	0	0.198695	0.158956	21
A5_schema posthoc_fields	0	0	0	0	0	0	0	21
A5_schema random_fields	0	0	0	0	0	0	0	21
A5_schema scrambled_fields	0	0	0	0	0	0	0	21

Table 11: Dataset-level AFCI contrasts against the structured matched-interface variant (A4_full).

Dataset	Comparator variant	Comparator AFCI	Structured AFCI	Structured minus comparator
ds000210_self_loop	A4_compressed	0.48889	0.494167	0.00527778
ds000210_self_loop	A4_posthoc	0	0.494167	0.494167
ds000210_self_loop	A4_scrambled	0	0.494167	0.494167
ds000210_self_loop	A5_long	0	0.494167	0.494167
ds000210_self_loop	scratchpad	0	0.494167	0.494167
ds000210_self_loop	A5_plain	0.178611	0.494167	0.315556
ds000210_self_loop	A5_schema	0	0.494167	0.494167
ds000210_self_loop	posthoc_fields	0	0.494167	0.494167
ds000210_self_loop	A5_schema	0	0.494167	0.494167
ds000210_self_loop	random_fields	0	0.494167	0.494167
ds000210_self_loop	A5_schema	0	0.494167	0.494167
ds000210_self_loop	scrambled_fields	0	0.494167	0.494167
ds000212_choice	A4_compressed	0.496528	0.503472	0.00694444
ds000212_choice	A4_posthoc	0	0.503472	0.503472
ds000212_choice	A4_scrambled	0	0.503472	0.503472
ds000212_choice	A5_long	0	0.503472	0.503472
ds000212_choice	scratchpad	0	0.503472	0.503472
ds000212_choice	A5_plain	0.176111	0.503472	0.327361
ds000212_choice	A5_schema	0	0.503472	0.503472
ds000212_choice	posthoc_fields	0	0.503472	0.503472
ds000212_choice	A5_schema	0	0.503472	0.503472
ds000212_choice	random_fields	0	0.503472	0.503472
ds000212_choice	A5_schema	0	0.503472	0.503472
ds000212_choice	scrambled_fields	0	0.503472	0.503472
ds001618_self_loop	A4_compressed	0.495694	0.491667	-0.00402778
ds001618_self_loop	A4_posthoc	0	0.491667	0.491667
ds001618_self_loop	A4_scrambled	0	0.491667	0.491667
ds001618_self_loop	A5_long	0	0.491667	0.491667
ds001618_self_loop	scratchpad	0	0.491667	0.491667
ds001618_self_loop	A5_plain	0.192708	0.491667	0.298958
ds001618_self_loop	A5_schema	0	0.491667	0.491667
ds001618_self_loop	posthoc_fields	0	0.491667	0.491667
ds001618_self_loop	A5_schema	0	0.491667	0.491667
ds001618_self_loop	random_fields	0	0.491667	0.491667
ds001618_self_loop	A5_schema	0	0.491667	0.491667
ds001618_self_loop	scrambled_fields	0	0.491667	0.491667
ds001882_choice	A4_compressed	0.514861	0.506667	-0.00819444
ds001882_choice	A4_posthoc	0	0.506667	0.506667
ds001882_choice	A4_scrambled	0	0.506667	0.506667
ds001882_choice	A5_long	0	0.506667	0.506667
ds001882_choice	scratchpad	0	0.506667	0.506667
ds001882_choice	A5_plain	0.191181	0.506667	0.315486
ds001882_choice	A5_schema	0	0.506667	0.506667
ds001882_choice	posthoc_fields	0	0.506667	0.506667
ds001882_choice	A5_schema	0	0.506667	0.506667
ds001882_choice	random_fields	0	0.506667	0.506667
ds001882_choice	A5_schema	0	0.506667	0.506667
ds001882_choice	scrambled_fields	0	0.506667	0.506667
ds004042_recall	A4_compressed	0.508889	0.510417	0.00152778
ds004042_recall	A4_posthoc	0	0.510417	0.510417
ds004042_recall	A4_scrambled	0	0.510417	0.510417
ds004042_recall	fields	0	0.510417	0.510417

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Table 11: Dataset-level AFCI contrasts against the structured matched-interface variant (A4_full). (continued)

Dataset	Comparator variant	Comparator AFCI	Structured AFCI	Structured minus comparator
ds004042_recall	A5_long	0	0.510417	0.510417
ds004042_recall	scratchpad	0.236944	0.510417	0.273472
ds004042_recall	A5_plain	0	0.510417	0.510417
ds004042_recall	posthoc_fields	0	0.510417	0.510417
ds004042_recall	A5_schema	0	0.510417	0.510417
ds004042_recall	random_fields	0	0.510417	0.510417
ds004042_recall	A5_schema	0	0.510417	0.510417
ds004917_uncertainty	scrambled_fields	0.496528	0.501389	0.00486111
ds004917_uncertainty	A4_compressed	0	0.501389	0.501389
ds004917_uncertainty	A4_posthoc	0	0.501389	0.501389
ds004917_uncertainty	fields	0	0.501389	0.501389
ds004917_uncertainty	A4_scrambled	0	0.501389	0.501389
ds004917_uncertainty	fields	0	0.501389	0.501389
ds004917_uncertainty	A5_long	0	0.501389	0.501389
ds004917_uncertainty	scratchpad	0.182847	0.501389	0.318542
ds004917_uncertainty	A5_plain	0	0.501389	0.501389
ds004917_uncertainty	A5_schema	0	0.501389	0.501389
ds004917_uncertainty	posthoc_fields	0	0.501389	0.501389
ds004917_uncertainty	A5_schema	0	0.501389	0.501389
ds004917_uncertainty	random_fields	0	0.501389	0.501389
ds004917_uncertainty	A5_schema	0	0.501389	0.501389
ds004917_uncertainty	scrambled_fields	0.494473	0.494473	0
narratives_formal	A4_compressed	0	0.494473	0
narratives_formal	A4_posthoc	0	0.494473	0.494473
narratives_formal	fields	0	0.494473	0.494473
narratives_formal	A4_scrambled	0	0.494473	0.494473
narratives_formal	fields	0	0.494473	0.494473
narratives_formal	A5_long	0	0.494473	0.494473
narratives_formal	scratchpad	0	0.494473	0.494473
narratives_formal	A5_plain	0.232462	0.494473	0.262011
narratives_formal	A5_schema	0	0.494473	0.494473
narratives_formal	posthoc_fields	0	0.494473	0.494473
narratives_formal	A5_schema	0	0.494473	0.494473
narratives_formal	random_fields	0	0.494473	0.494473
narratives_formal	A5_schema	0	0.494473	0.494473
narratives_formal	scrambled_fields	0	0.494473	0.494473

S4. Finite-Action-Code Behavioral Benchmark

The finite-action-code benchmark scores canonical final-action behavior under the finite action alphabet. Expected labels are scorer-side rule annotations derived from the condition map; the model-visible prompt specifies the task context, manipulation, module-lesion rule and allowed action ontology but not the hidden expected label used for scoring.

Table 12: Finite-action-code ontology and scorer-side expected labels.

Probe condition	Valid action codes	Expected before	Expected after	Interpretation
baseline	ACTION_A; ACTION_B	ACTION_A	ACTION_A	No specialized component is required; correct behavior is to keep ACTION_A.
reason_flip	ACTION_A; ACTION_B	ACTION_A	ACTION_B	The decisive reason is moved from A to B; reason-enabled variants should change.
memory_conflict	ACTION_A; RECALL_PRIOR; DEFER	ACTION_A	RECALL_PRIOR	A prior commitment conflicts with the immediate action impulse.
veto_cue	ACTION_A; VETO; DEFER	ACTION_A	VETO	A hard stop or constraint cue invalidates the first impulse.
self_continuity	ACTION_A; ACTION_B; DEFER	ACTION_A	ACTION_A	A persistent self-state or long-term commitment supports continuity.
irrelevant_cue	ACTION_A; ACTION_B	ACTION_A	ACTION_A	An irrelevant cue should not change the action.

Table 13: Finite-action-code behavioral benchmark criterion summary.

Criterion	Field	Value
finite-action-code behavioral benchmark	status	pass
finite-action-code behavioral benchmark	dataset_count	7
finite-action-code behavioral benchmark	scored records	57816
finite-action-code behavioral benchmark	source_scored records	61056
finite-action-code behavioral benchmark	planned calls	58896
finite-action-code behavioral benchmark	remote calls	58835
finite-action-code behavioral benchmark	unrecovered generations	1
finite-action-code behavioral benchmark	parse-error rate	1.700e-05
finite-action-code behavioral benchmark	unmapped action rate	0.00129722
finite-action-code behavioral benchmark	structured-control irrelevant false-positive rate	0.0083022
finite-action-code behavioral benchmark	structured-control behavioral composite > high-stochasticity datasets	7
finite-action-code behavioral benchmark	structured-control B_RSI > reason-lesion datasets	7
finite-action-code behavioral benchmark	structured-control B_VEI > veto-lesion datasets	7
finite-action-code behavioral benchmark	structured-control B_MCI > high-stochasticity datasets	7
finite-action-code behavioral benchmark	structured-control B_SCI > high-stochasticity datasets	7
finite-action-code behavioral benchmark	dataset-cluster bootstrap lower 95% bound	0.855732

Table 14: Behavioral component scores summarized by variant.

Variant	RSI μ	MCI μ	VEI μ	SCI μ	Cue μ	Comp. μ	FP μ	RSI σ	MCI σ	VEI σ	SCI σ	Cue σ	Comp. σ	FP σ	Sets
A4	0.973	0.773	0.989	0.990	0.931	0.931	0.008	0.023	0.151	0.019	0.021	0.037	0.037	0.019	7
A4_no reason	-0.044	0.788	0.954	0.954	0.663	0.663	0.044	0.113	0.089	0.112	0.113	0.102	0.102	0.113	7
A4_no veto	0.977	0.801	-0.008	0.990	0.690	0.690	0.009	0.019	0.081	0.019	0.021	0.020	0.020	0.019	7
A5	0.002	-0.001	-0.001	0.179	0.045	0.045	0.001	0.004	0.001	0.001	0.101	0.025	0.025	0.001	7

Table 15: Behavioral component scores by dataset namespace and variant.

Dataset	Variant	B_RSI	B_MCI	B_VEI	B_SCI	Cue	Composite	FP
ds000210_self_loop	A4	0.969444	0.808333	0.988889	0.994444	0.940278	0.940278	0.00277778
ds000210_self_loop	A4_no_reason	0	0.813889	0.997222	0.997222	0.702083	0.702083	0
ds000210_self_loop	A4_no_veto	0.969444	0.788889	-0.00277778	0.994444	0.6875	0.6875	0.00277778
ds000210_self_loop	A5	0	0	0	0.166667	0.0416667	0.0416667	0
ds000212_choice	A4	0.977778	0.555556	0.997222	1	0.882639	0.882639	0
ds000212_choice	A4_no_reason	-0.00277778	0.738889	0.994444	0.997222	0.681944	0.681944	0.00277778
ds000212_choice	A4_no_veto	0.988889	0.672222	0	1	0.665278	0.665278	0
ds000212_choice	A5	0	0	0	0.161111	0.0402778	0.0402778	0
ds001618_self_loop	A4	0.983333	0.816667	0.997222	1	0.949306	0.949306	0
ds001618_self_loop	A4_no_reason	-0.00277778	0.863889	0.997222	0.988889	0.711806	0.711806	0.00277778
ds001618_self_loop	A4_no_veto	0.983333	0.869444	0	1	0.713194	0.713194	0
ds001618_self_loop	A5	0.00277778	0	0	0.391667	0.0986111	0.0986111	0
ds001882_choice	A4	0.997222	0.852778	0.994444	0.997222	0.960417	0.960417	0.00277778
ds001882_choice	A4_no_reason	0	0.836111	1	1	0.709028	0.709028	0
ds001882_choice	A4_no_veto	1	0.847222	0	1	0.711806	0.711806	0
ds001882_choice	A5	0	0	0	0.0916667	0.0229167	0.0229167	0
ds004042_recall	A4	0.927171	0.907563	0.946779	0.943978	0.931373	0.931373	0.0504202
ds004042_recall	A4_no_reason	-0.29972	0.635854	0.70028	0.697479	0.433473	0.433473	0.29972
ds004042_recall	A4_no_veto	0.938375	0.879552	-0.0504202	0.943978	0.677871	0.677871	0.0504202
ds004042_recall	A5	0	0	0	0.159664	0.039916	0.039916	0
ds004917_uncertainty	A4	0.991667	0.908333	1	0.997222	0.974306	0.974306	0
ds004917_uncertainty	A4_no_reason	0	0.891667	1	1	0.722917	0.722917	0
ds004917_uncertainty	A4_no_veto	0.980556	0.838889	0	0.997222	0.704167	0.704167	0
ds004917_uncertainty	A5	0	0	0	0.0944444	0.0236111	0.0236111	0
narratives_formal	A4	0.964286	0.563492	0.996032	1	0.880952	0.880952	0
narratives_formal	A4_no_reason	0	0.738095	0.988095	1	0.681548	0.681548	0
narratives_formal	A4_no_veto	0.97619	0.710317	-0.00396825	0.996032	0.669643	0.669643	0.00396825
narratives_formal	A5	0.0119048	-0.00396825	-0.00396825	0.190476	0.0486111	0.0486111	0.00396825

Table 16: Predefined dataset-direction behavioral criterion contrasts.

Dataset	structured comp.	reason-lesion comp.	veto-lesion comp.	HS comp.	structured-HS comp.	reason contrast	veto contrast	memory contrast	self contrast
ds000210_self_loop	0.940278	0.702083	0.6875	0.0416667	0.898611	0.969444	0.991667	0.808333	0.827778
ds000212_choice	0.882639	0.681944	0.665278	0.0402778	0.842361	0.980556	0.997222	0.555556	0.838889
ds001618_self_loop	0.949306	0.711806	0.713194	0.0986111	0.850694	0.986111	0.997222	0.816667	0.608333
ds001882_choice	0.960417	0.709028	0.711806	0.0229167	0.9375	0.997222	0.994444	0.852778	0.905556
ds004042_recall	0.931373	0.433473	0.677871	0.039916	0.891457	1.22689	0.997199	0.907563	0.784314
ds004917_uncertainty	0.974	0.723	0.704	0.024	0.951	0.992	1	0.908	0.903
narratives_formal	0.880952	0.681548	0.669643	0.0486111	0.832341	0.964286	1	0.56746	0.809524

S5. Strong Stochastic Controls and Entropy Diagnostics

Strong stochastic controls test whether alternative stochastic implementations recover the structured-control profile. Entropy diagnostics summarize action variability under raw strings, rule-canonical actions, semantic clusters and action-family mappings. The calibration tables document that the pre-defined entropy-closeness criterion was not achieved.

Table 17: Proxy token/compute accounting summary. Provider-side prompt-token, completion-token and total-token usage were not uniformly retained; this table reports availability and proxy measurements for length and structure.

Audit item	Measurement or proxy	Availability or value	Evidence source	Interpretation
provider token usage	prompt_tokens, completion_tokens, total_tokens	not uniformly retained	token_verbosity_audit criterion summary reports provider token usage unavailable	provider-token accounting incomplete
prompt length	model-visible prompt character or token count	not uniformly retained	no complete prompt-length column in retained token audit summary	prompt-token matching unavailable
completion length proxy	json_trace_chars, final_action_chars, first_impulse_chars	84,516 trace rows; 552 source/dataset/variant summaries	retained token-verbosity audit table	field-length/verbosity proxy only
trace structure proxy	reason_node_count, branch_count, memory_item_count	84,516 trace rows; 552 summaries	retained token-verbosity audit table	structure proxy for compute-risk assessment
schema field count	matched-interface emitted JSON schema	nine external fields in matched-interface protocol	matched-interface protocol schema	controls schema availability under the matched interface
retry count	provider retry attempts	not uniformly retained	unrecovered generations are reported separately by experiment	retry-cost inference unavailable
latency proxy	latency_ms	available where retained in token_verbosity_audit	wall-clock proxy affected by provider and network conditions	indirect model-compute/cost proxy
compressed structured-control sensitivity	AFCI_mean	A4_full=0.500; A4_compressed=0.499	matched-interface AFCI table	coupling persisted under compressed trace expression
A5_long scratchpad control	AFCI_mean	A5_long scratchpad=0.000	matched-interface AFCI table	compute matching not established

Table 18: Robustness and stochastic-control criterion summary. Prompt-paraphrase, expanded counterfactual and placebo rows are auxiliary robustness blocks. They are distinct from the formal counterfactual-transform quality-control screen and the consistency-passing transform reported in Supplementary Table 33.

Experiment	Status	Calls	Remote	Unrec.	Unrec.	Min ds	HS rec.	AFCI ds	Parse n	Parse r	Kappa	Leak
strong stochastic controls	pass	15490	15490	0	0	7	0	-	-	-	-	-
A5_empirical	predefined	17082	17082	0	0	7	0	-	-	-	-	-
entropy_attempt	closeness not satisfied											
prompt	pass	11400	11400	0	0	17	0	-	-	-	-	-
paraphrases	pass											
expanded counterfactual	auxiliary	30584	30584	0	0	61	0	-	-	-	-	-
probes	pass											
placebo	pass	9960	9960	0	0	24	0	-	-	-	-	-
components	pass											
cross-model subset	pass	-	-	-	-	-	-	-	-	-	-	-
open-weight	pass	-	-	-	-	-	-	7	8	0.000762	-	-
reference model	pass	-	-	-	-	-	-	-	-	-	-	-
expanded blinded	pass	-	-	-	-	-	-	-	-	-	0.961	0
audit												

Table 19: Entropy diagnostics summarized by experiment source and variant.

Source	Variant	Raw H mean	Rule H mean	Semantic H mean	Family H mean	Raw H SD	Rule H SD	Semantic H SD	Family H SD	Aggregate rows
counterfactual probes	A4	4.180	2.647	3.962	0.863	1.703	1.490	1.625	0.461	70
	A4_no reason									
counterfactual probes	A5	4.013	2.549	3.851	0.883	1.753	1.543	1.683	0.512	70
	A4_no veto									
counterfactual probes	A4	4.179	2.591	3.969	0.913	1.734	1.508	1.709	0.490	70
	A5	5.519	3.362	5.445	1.086	1.410	1.738	1.432	0.452	70
A5_emp.										
entropy	A4	6.596	5.106	6.360	1.070	1.292	1.541	1.301	0.411	21
A5_emp.										
entropy	A5	7.569	5.600	7.514	1.241	0.956	1.390	0.979	0.131	21
A5_emp.	A5_emp. entropy									
entropy	A4	7.584	5.477	7.529	1.236	0.924	1.432	0.951	0.134	21
	A4_no	4.445	3.443	4.312	0.822	1.594	1.628	1.546	0.393	33
placebo_components										
	A4_no reason	4.156	3.232	4.027	0.879	1.622	1.472	1.596	0.500	33
	A4_no									
placebo_components	veto	4.698	3.546	4.551	0.865	1.508	1.678	1.460	0.432	33
	A5	5.364	4.048	5.330	1.091	1.442	1.677	1.450	0.378	33
placebo_components										
	A4	5.232	3.848	5.000	1.159	1.409	1.501	1.341	0.466	21
	A4_no reason									
prompt_paraphrases		4.895	3.419	4.720	1.103	1.503	1.495	1.458	0.514	21
	A4_no									
prompt_paraphrases	veto	5.464	4.010	5.267	1.002	1.356	1.515	1.326	0.410	21
	A5	6.302	4.653	6.183	1.204	0.995	1.340	1.000	0.251	21
prompt_paraphrases										
strong stochastic										
controls	A4	7.009	5.462	6.777	1.090	1.849	1.757	1.815	0.415	7
strong stochastic										
controls	A5	7.998	5.914	7.943	1.204	1.377	1.545	1.401	0.228	7
strong stochastic	compute									
controls	control	7.994	5.915	7.936	1.238	1.422	1.569	1.452	0.151	7
strong stochastic	A5_entropy									
controls	attempt	7.941	5.896	7.896	1.161	1.465	1.754	1.463	0.211	7
strong stochastic	A5_self									
controls	consistency	7.994	5.926	7.928	1.173	1.382	1.640	1.414	0.176	7

Table 20: Predefined empirical entropy-attempt calibration criterion summary.

Criterion	Field	Value
strict_a5_em_calibration	calibration_events_per_dataset	50
strict_a5_em_calibration	closeness_threshold	0.15
strict_a5_em_calibration	grid_coverage_status	complete
strict_a5_em_calibration	max_selected_strict_entropy_gap	1.17146
strict_a5_em_calibration	note	raw/canonical/semantic/action-family closeness did not meet threshold
strict_a5_em_calibration	observed_required_setting_count	28
strict_a5_em_calibration	required_setting_count	28
strict_a5_em_calibration	selected_temperature	0.2
strict_a5_em_calibration	selected_top_p	0.75
strict_a5_em_calibration	selected_variant	A5_attempt_t020_p075
strict_a5_em_calibration	status	predefined closeness not satisfied
strict_a5_em_calibration	strict_entropy_close_enough	False
strict_a5_em_calibration	wording_rule	Entropy matching is reported only when the predefined closeness criterion is satisfied; here the empirical attempt did not satisfy that criterion.

Table 21: Predefined empirical entropy-attempt calibration grid.

Variant	Temp.	Top p	Raw H gap	Rule H gap	Semantic H gap	Family H gap	Strict score	Max strict gap
A5_attempt	0.2	0.6	0.944	1.300	1.096	0.302	0.910	1.300
t020_p060	0.2	0.75	0.973	1.171	1.110	0.353	0.902	1.171
A5_attempt	0.2	0.9	1.102	1.420	1.224	0.404	1.037	1.420
t020_p075	0.2	1	0.955	1.331	1.094	0.365	0.936	1.331
A5_attempt	0.4	0.6	1.125	1.545	1.271	0.388	1.082	1.545
t040_p060	0.4	0.75	1.154	1.462	1.295	0.249	1.040	1.462
A5_attempt	0.4	0.9	1.070	1.328	1.217	0.362	0.994	1.328
t040_p075	0.4	1	1.186	1.334	1.314	0.317	1.038	1.334
A5_attempt	0.6	0.6	1.129	1.518	1.264	0.296	1.052	1.518
t060_p060	0.6	0.75	1.123	1.373	1.260	0.403	1.040	1.373
A5_attempt	0.6	0.9	1.114	1.492	1.253	0.504	1.091	1.492
t060_p075	0.6	1	1.155	1.520	1.273	0.303	1.063	1.520
A5_attempt	0.8	0.6	1.167	1.539	1.295	0.371	1.093	1.539
t080_p060	0.8	0.75	1.043	1.344	1.178	0.322	0.972	1.344
A5_attempt	0.8	0.9	1.068	1.414	1.207	0.324	1.003	1.414
t080_p075	0.8	1	1.058	1.704	1.205	0.307	1.069	1.704
A5_attempt	1	0.6	0.904	1.305	1.026	0.418	0.913	1.305
t100_p060	1	0.75	0.973	1.476	1.106	0.469	1.006	1.476
A5_attempt	1	0.9	0.960	1.403	1.099	0.404	0.966	1.403
t100_p075	1	1	1.017	1.438	1.164	0.394	1.003	1.438
A5_attempt	1.2	0.6	1.045	1.579	1.184	0.435	1.061	1.579
t120_p060	1.2	0.75	1.043	1.136	1.188	0.577	0.986	1.188
A5_attempt	1.2	0.9	0.984	1.389	1.128	0.395	0.974	1.389
t120_p075	1.2	1	1.029	1.298	1.170	0.469	0.991	1.298
A5_attempt	1.4	0.6	1.041	1.357	1.179	0.525	1.025	1.357
t140_p060	1.4	0.75	1.028	1.419	1.157	0.397	1.000	1.419
A5_attempt	1.4	0.9	1.051	1.357	1.198	0.437	1.011	1.357
t140_p075	1.4	1	0.966	1.424	1.119	0.439	0.987	1.424

Table 22: Finite-action-code entropy-calibration criterion summary.

Criterion	Field	Value
finite-action entropy calibration	structured-control behavioral composite > selected high-stochasticity datasets	7
finite-action entropy calibration	behavior pass threshold	6
finite-action entropy calibration	calibration events per dataset	14
finite-action entropy calibration	calibration maximum predefined entropy gap	1.22474
finite-action entropy calibration	calibration mean predefined entropy gap	1.13035
finite-action entropy calibration	calls	11088
finite-action entropy calibration	closeness threshold	0.15
finite-action entropy calibration	dataset count	7
finite-action entropy calibration	dry run	False
finite-action entropy calibration	experiment	finite-action entropy calibration
finite-action entropy calibration	unrecovered-generation rate	0
finite-action entropy calibration	grid setting count	14
finite-action entropy calibration	maximum model-generation budget	12000
finite-action entropy calibration	holdout events per dataset	18
finite-action entropy calibration	holdout maximum predefined entropy gap	1.34765
finite-action entropy calibration	holdout mean predefined entropy gap	1.14711
finite-action entropy calibration	holdout predefined entropy close enough	False
finite-action entropy calibration	holdout_unmapped action rate	0
finite-action entropy calibration	planned calibration calls	8820
finite-action entropy calibration	planned calls	11088
finite-action entropy calibration	planned holdout calls	2268
finite-action entropy calibration	remote calls	11088
finite-action entropy calibration	selected high-stochasticity candidate did not recover structured-control behavior	True
finite-action entropy calibration	selected behavior bootstrap lower 95% bound	0.863095
finite-action entropy calibration	selected temperature	0.2
finite-action entropy calibration	selected top-p	0.7
finite-action entropy calibration	selected variant	A5_attempt_t02_p07
finite-action entropy calibration	status	predefined closeness not satisfied
finite-action entropy calibration	unrecovered generations	0
finite-action entropy calibration	wording rule	Entropy calibration is reported as matched only when holdout closeness and behavioral criteria are satisfied; here the retained attempt did not satisfy the required criteria.

Table 23: Finite-action-code entropy-calibration grid.

Variant	Sets	Rows	Mean gap	Max gap	Unmap	Raw H	Code H	Family H	Temp.	Top p
A5_attempt	7	588	1.138	1.189	0	1.138	1.138	1.138	0.1	0.7
t01_p07	7	588	1.171	1.249	0.002	1.171	1.171	1.171	0.1	1
A5_attempt	7	588	1.130	1.225	0.002	1.130	1.130	1.130	0.2	0.7
t02_p07	7	588	1.142	1.250	0	1.142	1.142	1.142	0.2	1
A5_attempt	7	588	1.139	1.189	0	1.139	1.139	1.139	0.4	0.7
t04_p07	7	588	1.164	1.219	0	1.164	1.164	1.164	0.4	1
A5_attempt	7	588	1.162	1.250	0.002	1.162	1.162	1.162	0.6	0.7
t06_p07	7	588	1.161	1.223	0	1.161	1.161	1.161	0.6	1
A5_attempt	7	588	1.157	1.225	0.002	1.157	1.157	1.157	0.8	0.7
t08_p07	7	588	1.156	1.223	0	1.156	1.156	1.156	0.8	1
A5_attempt	7	588	1.153	1.223	0	1.153	1.153	1.153	1	0.7
t10_p07	7	588	1.153	1.284	0.003	1.153	1.153	1.153	1	1
A5_attempt	7	588	1.152	1.223	0	1.152	1.152	1.152	1.2	0.7
t12_p07	7	588	1.142	1.219	0.002	1.142	1.142	1.142	1.2	1

Table 24: Finite-action-code entropy-calibration holdout grid.

Variant	Sets	Rows	Mean gap	Max gap	Unmap	Raw H	Code H	Family H	Temp.	Top p
A5	7	756	1.165	1.299	0	1.165	1.165	1.165	–	–
A5_attempt	7	756	1.147	1.348	0	1.147	1.147	1.147	0.2	0.7

S6. Robustness, Open-Weight Reference Model and Blinded Audit

Robustness checks evaluate the direction of the result across tested prompts, tested model identifiers, one local open-weight reference model and blinded annotation. These checks support robustness across the tested settings; they do not imply coverage of untested models or prompts. The agreement table reports Cohen’s κ , a chance-adjusted nominal-agreement statistic (Cohen, 1960).

Table 25: Cross-model subset criterion summary.

Model	Status	Exp.	Profile	Calls	Remote	Unrecov.	Unrec. rate	Vars.	Ctrls.	Min ctrl ds	Strong HS ds	Fallback	SC>HS ds
gpt-5.4 mini	pass	cross subset	full	1995	1995	0	0	5	4	5	0	qwen3 30b-a3b	7
	pass	cross subset	full	1995	1995	0	0	5	4	6	0	qwen3 30b-a3b	7
deepseek-chat	pass	cross subset	full	1995	1995	0	0	5	4	6	0	qwen3 30b-a3b	7
	pass	cross subset	full	1995	1995	2	0.001	5	4	7	0	qwen3 30b-a3b	7

Table 26: Open-weight Qwen2.5-14B-Instruct AFCI summary by variant.

Variant	R μ	M μ	Veto μ	Self μ	Supp. μ	AFCI μ	AFCI+ μ	R σ	M σ	Veto σ	Self σ	Supp. σ	AFCI σ	AFCI+ σ	Rows
A4	1	0.5	1	1	1	0.875	0.9	0	0	0	0	0	0	0	21
A4_no reason	0.5	0.5	1	1	1	0.75	0.8	0	0	0	0	0	0	0	21
A4_no veto	1	0.5	0.5	1	1	0.75	0.8	0	0	0	0	0	0	0	21
A5	0.446	0.25	0.5	0.393	0	0.397	0.318	0.019	0	0	0.031	0	0.008	0.006	21
A5_schema posthoc_fields	0	0	0	0	0	0	0	0	0	0	0	0	0	0	21

Table 27: Expanded blinded audit reliability by annotation dimension.

Dimension	Annotator A	Annotator B	Overlap items	Agreement	Cohen kappa
final action reason-consistent	annotator_01	annotator_02	1200	0.958333	0.912127
final action reason-consistent	annotator_01	annotator_03	1200	0.955833	0.907158
final action reason-consistent	annotator_02	annotator_03	1200	0.990833	0.980522
action respects relevant	annotator_01	annotator_02	1200	0.997500	0.989211
memory					
action respects relevant	annotator_01	annotator_03	1200	0.997500	0.989211
memory					
action respects relevant	annotator_02	annotator_03	1200	1.000000	1.000000
memory					
veto_was_appropriate	annotator_01	annotator_02	1200	1.000000	1.000000
veto_was_appropriate	annotator_01	annotator_03	1200	1.000000	1.000000
veto_was_appropriate	annotator_02	annotator_03	1200	1.000000	1.000000
veto changed first impulse	annotator_01	annotator_02	1200	1.000000	1.000000
veto changed first impulse	annotator_01	annotator_03	1200	1.000000	1.000000
veto changed first impulse	annotator_02	annotator_03	1200	1.000000	1.000000
reason_looks_posthoc	annotator_01	annotator_02	1200	0.960000	0.917801
reason_looks_posthoc	annotator_01	annotator_03	1200	0.961667	0.921303
reason_looks_posthoc	annotator_02	annotator_03	1200	0.995000	0.989652
verbose/template-like	annotator_01	annotator_02	1200	0.998333	0.971364
without control					
verbose/template-like	annotator_01	annotator_03	1200	0.996667	0.942728
without control					
verbose/template-like	annotator_02	annotator_03	1200	0.998333	0.972113
without control					
irrelevant cue caused	annotator_01	annotator_02	1200	0.974167	0.865842
inappropriate change					
irrelevant cue caused	annotator_01	annotator_03	1200	0.972500	0.860015
inappropriate change					
irrelevant cue caused	annotator_02	annotator_03	1200	0.993333	0.965260
inappropriate change					

Annotation ethics scope: three trained annotators contributed output-level quality-control labels under a fixed rubric and full-overlap blinded design. The audit did not collect personal, behavioral, health, intervention or neural data from annotators or any other individuals. Annotation was performed as expert quality-control scoring of model outputs for the project record rather than as human-subjects data collection.

S7. Validation Source-Data Map, Reproducibility and Export Coverage

The reproducibility map links each claim-supporting statement to its table, result export, criterion summary, analysis entry point and checksum. Full SHA-256 values are retained in the corresponding CSV export; the PDF table reports the first 12 hexadecimal characters for readability. The validation source-data map below gives the direct source-data and script paths for the main validation figure, the SWE-bench Lite reliability displays and the formal control definitions.

Table 28: Validation source-data and control-construction map.

Item	Role	Source data, config or script path	Checksum or provenance
Main validation panel a source data	Open-weight core no-context controls	manuscript/figures/source_data/figure5a_open_weight_core_controls.csv	12ef8d181136
Main validation panel b source data	Open-weight state-boundary lesion	manuscript/figures/source_data/figure5b_open_weight_state_boundary_lesion.csv	803140cd02ce
Main validation panel c source data	SWE-bench Lite full 300-record incremental AUC	manuscript/figures/source_data/figure5c_swebench_incremental_auc.csv	261436e42398
Main validation panel d source data	SWE-bench Lite single-predictor AUC comparison	manuscript/figures/source_data/figure5d_swebench_single_predictor_auc.csv	8897f8c61c79
Main validation panel e source data	SWE-bench Lite sensitivity and binding-guard summary	manuscript/figures/source_data/figure5e_swebench_sensitivity_guard.csv	4dc490ac380e
Main validation rendering and export QA	Figure rendering and export-QA script	manuscript/figures/scripts/build_validation_figure5.R;	61d69f9d8264 /
SWE-bench reliability calibration source data	Issue-grouped five-fold reliability calibration for the full 300-record issue-to-file endpoint	audit/figure5_primary_validation_export_qc_20260530.csv manuscript/figures/source_data/extended_data_figure9_swebench_reliability_calibration.csv; manuscript/figures/source_data/swebench_cv_reliability_predictions.csv	80f8d170bb37 archived manifest
SWE-bench threshold analysis source data	Thresholded localization-guard analysis over predicted hit@3 probabilities	manuscript/figures/source_data/extended_data_figure10_swebench_threshold_analysis.csv;	archived manifest
Main Table 1 source data	Claim ledger with supported claims and excluded inferences	manuscript/figures/scripts/build_swebench_reliability_figures.R manuscript/tables/table1_validation_evidence_interpretation.csv	archived manifest
Supplementary Table 33 source data	Validation controls and interpretation status	manuscript/tables/supplementary_table_33_validation_controls.csv	7a88ca507fb7
Decisive-field positive-control summaries	Two-model decisive-field and wrapper result tables	manuscript/tables/table8_decisive_field_positive_control_condition_summary.csv; manuscript/tables/table8_control_binding_wrapper_scrambled_summary.csv	archived manifest
Decisive-field positive-control leakage audit	Prompt/scorer leakage and scope checks	manuscript/tables/table8_decisive_field_positive_control_leakage_audit.csv	e1dfdc5005ce
Local strict matching summaries	Local Qwen2.5-1.5B forced-budget entropy/token audit	manuscript/tables/table16_local_strict_matching_condition_summary.csv;	archived manifest
SWE-bench Lite issue-to-file summaries	Complete 300-record issue-localization summaries and execution-slice preparation status	manuscript/tables/table16_local_strict_matching_gate_summary.csv results/paper1_revision/real_task_predictive_validity/swebench_v2_full300_api_20260529_merged/gate_status.json; results/paper1_revision/real_task_predictive_validity/swebench_v2_full300_api_20260529_merged/official_execution_slice_environment_status.json; results/paper1_revision/real_task_predictive_validity/swebench_v2_full300_api_20260529_merged/official_execution_slice_analysis_v3_apply_repair/official_execution_slice_gate_status.json	archived manifest
Open-weight matrix config path	Model, scaffold, task-family and variant manifest	results/paper1_revision/remote_5090_run3_20260508/mmi_formal_execution_manifest.json	11e3390109fd
Formal API validation config path	API validation run manifest	results/paper1_revision/remote_5090_run3_20260508/formal_api_validation/formal_api_full_20260510_0030/run_manifest.json	8712d8ea8b1b
Distribution-matched control construction	Offline/no-context action-prior policy and prompt construction	src/freewill/structured_control_validation.py	eba9d1039034

Continued on next page

Table 28: Validation source-data and control-construction map. (continued)

Item	Role	Source data, config or script path	Checksum or provenance
Strict-lesion construction	Removal of decisive intervention-carrying field with non-decisive context retained	src/freewill/structured_control_validation.py	e9a9d1039034
Counterfactual transform QC and consistency-passing transform	Transform definitions and formal holdout launcher	src/freewill/structured_control_validation.py; scripts/run/run_formal_open_weight_validation.py	e9a9d1039034 / 0f9cfd41e1af
Token/compute criterion script	Prompt-token, total-token and latency-proxy criterion analysis	scripts/run/analyze_formal_validation_results.py;	404f5edb4132 /
Entropy-prior behavioral-control script	Calibration-split action-prior selection and entropy-prior no-field behavioral control	scripts/run/audit_open_weight_token_entropy.py	f037f6424513
Local strict matching script	Local tokenizer padding, fixed-generation and forced-budget audit implementation	scripts/run/select_formal_entropy_settings.py;	f29b12ebe627 /
SWE-bench Lite localization scripts	Dataset fetch, oracle-free issue-to-file prompts, shard merge, readiness audit, execution-slice patch generation and v3 apply-check/repair filtering	src/freewill/structured_control_validation.py scripts/run/run_strict_matching_local.py; src/freewill/strict_matching_local.py scripts/run/run_swebench_file_localization_v2_api.py; scripts/run/merge_swebench_fileloc_v2_shards.py; scripts/run/run_swebench_execution_slice_patch_generation_api.py; scripts/run/run_swebench_patch_generation_v3_apply_repair.py; src/freewill/real_task_predictive_validity.py	e9a9d1039034 archived manifest archived manifest

Table 29: Claim-to-evidence reproducibility map.

Claim	Main table	Result export	Criterion-summary alias	Analysis entry point	Checksum prefix
baseline lesion matrix separates unpredictability and trace structure	Table 2	table2_baseline lesion_matrix gate_summary.csv	gate/baseline_lesion_matrix gate_status.json	paper1-run-baseline-lesion-matrix configs/ai_matrix.yaml	5f2fae5d2ab5
matched-interface AFCI rules out field availability alone	Table 3	table3_schema_gate_summary.csv table3_schema_afci_by_variant.csv table3_schema_comparator_gate_status.csv	gate/matched_interface_controls gate_status.json	paper1-run-matched-interface-controls configs/schema_equalized_controls.yaml	2fdaffdbfdb
finite-action-code behavior supports action-level component coupling	Table 4	table4_behavior_gate_summary.csv table4_behavior_dataset_gate_contrasts.csv	gate/finite_action_behavior gate_status.json	paper1-run-finite-action-behavior configs/behavioral_counterfactuals.yaml	55820262e805
strong stochastic controls do not recover structured-control profile	Table 9	table7_robustness gate_summary.csv	gate/strong_stochastic_controls gate_status.json	paper1-run-strong-stochastic-controls configs/ai_matrix.yaml	1f1de7d4d18d
predefined entropy closeness was not satisfied	Table 6 Table 12	table6_strict_a5_em_calibration_gate_summary.csv table6_a5_entropy_matching_calibration_gate_summary.csv table12_entropy_calibration_threshold_summary.csv	gate/entropy_calibration_diagnostic gate_status.json	paper1-run-entropy-diagnostics paper1-run-a5-entropy-calibration	b6150257 c3ae
cross-model tested subset preserves structured-control greater-than high-stochasticity direction	Table 9	table7_cross_model gate_summary.csv	gate/cross_model_subset gate_status.json	paper1-run-cross-model-subset configs/ai_matrix.yaml	2f8b5ff04901
open-weight reference model preserves structured-control greater-than high-stochasticity AFCI direction	Table 9 / Table 10	table7_open_weight_afci_by_variant.csv table7_open_weight_afci_by_dataset_variant_seed.csv	gate/open_weight_anchor gate_status.json	paper1-run-open-weight-anchor configs/open_weight_anchor.yaml	d76affc4ffc8
decisive-field positive control recovers the finite-action binding readout	Table 8	table8_decisive_field positive_control_condition_summary.csv	decisive_field positive_control/gate_status.json	scripts/run run_decisive_field positive_control_api.py	archived manifest
local wrapper reduces scrambled-field binding	Table 8	table8_control_binding_wrapper scrambled_summary.csv	decisive_field positive_control/gate_status.json	scripts/run run_decisive_field positive_control_api.py	archived manifest
bounded local forced-budget entropy/token audit preserves structured-state advantage	Table 16	table16_local_strict_matching condition_summary.csv	local_strict_matching gate_status.json	scripts/run run_strict_matching_local.py	0206bfa8d6db
SWF-bench Lite issue-to-file validation passes oracle-free predictor gate	Table 17	swebench_v2_full300 api_20260529_merged/gate_status.json	real_task_predictive_validity swebench_v2_readiness_audit.json	scripts/run run_swebench_file_localization_v2_api.py	archived manifest
blinded audit supports scorer interpretation	Table 9	table7_blind_audit_reliability by_dimension.csv	gate/strict_blind_audit gate_status.json	paper1-run-blind-audit-export configs/ai_matrix.yaml	bffe01d6d1fd

Interface-diagnostic rows are summarized separately from effect estimates. The taxonomy distinguishes final-criterion-compatible interface diagnostics, non-final provider-interface attempts and entropy-calibration diagnostics. The export-coverage table lists the table exports underlying the manuscript and this supplement.

Table 30: Interface-diagnostic taxonomy and interpretation.

Source	Experiment	Model	Diagnostic type	Rows	Interpretation
Cross-model robustness	Gemini full run	gemini-2.5-flash	payload not object	2	Final criterion met; retained as interface diagnostic.
Provider-interface compatibility check	Qwen direct-provider check	qwen3-30b-a3b	HTTP 400	100	Compatibility rows outside the final-evidence criteria.
Entropy-calibration diagnostic	entropy-attempt calibration	qwen3-30b-a3b fallback	HTTP 500 / fallback metadata	594	Diagnostic calibration rows; predefined closeness was not satisfied.
Baseline lesion matrix	aggregate baseline accounting	gpt-5.4-mini	aggregate only	16	Aggregate unrecovered-output count; no trace-level taxonomy.

Table 31: Table-export coverage. Full-precision values are retained in the CSV exports. Data rows exclude the header row and any empty trailing lines.

Export label	Data rows	Columns	Leading columns
Variant name registry	17	5	canonical variant, used in, definition, aliases, role
Mechanistic claim map	3	5	claim component, operational test, contrary result, interpretive boundary
Baseline lesion-matrix criterion summary	12	3	criterion, field, value
Baseline trace metrics by dataset, variant and replicate index	168	19	variant, VBI, SCI, VEI, RSI, ACI, CSVI, unpredictability ...
Baseline trace metrics by variant	8	18	variant, VBI mean, SCI mean, VEI mean, RSI mean, ACI mean, CSVI mean, unpredictability mean ...
Matched-interface AFCI by dataset, variant and replicate index	189	10	dataset namespace, variant, replicate index, reason alignment, memory alignment, veto alignment ...
Matched-interface AFCI by variant	9	16	variant, alignment means, AFCI mean, AFCI with suppression mean, SD fields ...
Matched-interface dataset contrasts	56	5	dataset namespace, variant, AFCI, structured full AFCI, paired contrast
Matched-interface comparator criterion status	7	4	control variant, dataset count, structured greater-than-control dataset count, recovery count
Matched-interface criterion summary	11	3	criterion, field, value
Finite-action condition rates	504	7	dataset namespace, variant, replicate index, condition, correct rate, unmapped rate, row count
Finite-action dataset criterion contrasts	7	10	dataset namespace, structured variant, ablation comparators, high-stochasticity comparator, component contrasts ...
Finite-action criterion summary	16	3	criterion, field, value
Finite-action metrics by dataset, variant and replicate index	84	16	dataset namespace, variant, replicate index, condition rates and component scores ...
Finite-action summary by dataset and variant	28	9	dataset namespace, variant, B scores, cue specificity, behavioral structured action control ...
Finite-action summary by variant	4	16	variant, component-score means, behavioral composite mean, false-positive and row-count summaries ...
Entropy diagnostics by source and variant	20	11	source, variant, raw, canonical, semantic and action-family entropy means and SDs ...
Full entropy diagnostic units	594	14	source, output namespace, variant, replicate index, row count, unique action counts ...
Strict entropy-attempt calibration grid	28	12	variant, temperature, top-p, paired counts, absolute entropy gaps ...
Strict entropy-attempt criterion summary	13	3	criterion, field, value
Finite-action entropy-calibration grid	14	11	variant, dataset count, row count, predefined entropy gap, unmapped action rate, entropy gaps ...
Finite-action entropy-calibration criterion summary	30	3	criterion, field, value
Finite-action entropy-calibration holdout grid	2	11	variant, dataset count, row count, predefined entropy gap, unmapped action rate, entropy gaps ...
Blinded-audit reliability by dimension	7	6	dimension, annotator labels, overlap items, agreement, Cohen's kappa
Cross-model criterion summary	4	14	model, status, experiment profile, calls, interface diagnostics, directional dataset counts ...
Open-weight AFCI by dataset, variant and replicate index	105	10	dataset namespace, variant, replicate index, alignment terms and AFCI

Continued on next page

Table 31: Table-export coverage. Full-precision values are retained in the CSV exports.
Data rows exclude the header row and any empty trailing lines. (continued)

Export label	Data rows	Columns	Leading columns
Open-weight APCI by variant	5	16	variant, alignment means, APCI means, SD fields and rows
Robustness criterion summary	8	13	experiment, status, calls, interface diagnostics, directional counts, parse and audit statistics
Decisive-field positive-control condition summary	12	13	model, condition, rows, accuracy, wrapped accuracy, wrapper change rate ...
Decisive-field positive-control gate summary	2	10	model, full-state accuracy, only-decisive accuracy, best control accuracy, recovery fraction ...
Decisive-field positive-control task cells	240	4	model, task family, full-state accuracy, only-decisive accuracy
Decisive-field positive-control leakage audit	5	4	audit item, evidence, status, interpretation
Control-binding wrapper scrambled summary	2	7	model, rows, raw scrambled following, wrapped scrambled following, raw expected accuracy ...
Local strict matching condition summary	2	10	condition, rows, accuracy, invalid rate, canonical-action entropy, token and latency means
Local strict matching gate summary	1	12	status, prompt match rate, completion match rate, total-token ratio, latency ratio ...
SWE-bench file-localization predictor summary	11	5	predictor, hit@3 AUC, hit@3 Pearson, no-violation AUC, sign convention
SWE-bench file-localization model summary	6	13	model, model-instance rows, localization hit@3, violation rates, binding score and execution-slice boundary columns
SWE-bench file-localization gate summary	1	21	status, rows, model-instance rows, hit@3 counts, AUC summaries, wrapper status and execution-slice boundary columns
Derived effect-size summary	6	3	evidence block, direction, stronger summary
Proxy token/compute audit summary	9	5	audit item, measurement or proxy, availability, evidence source, interpretation
Entropy calibration threshold summary	44	12	source, variant, entropy gaps, strict threshold, pass flag
Interface-diagnostic taxonomy	4	6	source, experiment, model, diagnostic type, rows, interpretation
Claim-to-evidence reproducibility map	12	7	claim, main table, result export, criterion summary, analysis entry point, SHA-256
Statistical sensitivity summary	8	10	contrast, independent unit, direction count, minimum contrast, sign test

S8. Statistical and Scoring Definitions

Table 32: Dataset-level statistical sensitivity summary for primary contrasts. Exact sign tests are one-sided under a 0.5 null probability for a positive dataset direction; two-sided values are doubled and capped at one.

Contrast	Unit	Units	Positive units	Minimum contrast	Mean contrast	LODO minimum mean	Top-effect removed mean	One-sided sign p
finite behavior: structured-control-high-stochasticity composite	dataset namespace	7	7	0.832	0.886	0.875	0.875	0.0078125
reason ablation: structured-control-reason lesion B_RSI	dataset namespace	7	7	0.964	1.017	0.982	0.982	0.0078125
veto ablation: structured-control-veto lesion B_VEI	dataset namespace	7	7	0.992	0.997	0.996	0.996	0.0078125
memory behavior: structured-control-high-stochasticity B_MCI	dataset namespace	7	7	0.556	0.774	0.751	0.751	0.0078125
self-continuity behavior: structured-control-high-stochasticity B_SCI	dataset namespace	7	7	0.608	0.811	0.795	0.795	0.0078125
schema APCI: structured full-stochastic plain	dataset namespace	7	7	0.262	0.302	0.297	0.297	0.0078125
schema APCI: structured full-weakest control comparator	dataset namespace	7	7	0.492	0.500	0.499	0.499	0.0078125
open-weight APCI: structured-control-high-stochasticity	dataset namespace	7	7	0.467	0.478	0.476	0.476	0.0078125

Let $\mathcal{D}$ denote the seven dataset namespaces and $\mathcal{I}_{d,v}$ the retained generations for dataset d and variant v . Raw final-action unpredictability is reported as Shannon entropy (Shannon, 1948),

$$H_{d,v} = - \sum_{a \in \mathcal{A}_{d,v}} \hat{p}_{d,v}(a) \log_2 \hat{p}_{d,v}(a), \quad (1)$$

where $\mathcal{A}_{d,v}$ is the observed final-action support and $\hat{p}_{d,v}$ is the empirical action frequency. The baseline structured composite is

$$C_{d,v} = \frac{1}{4} (\text{SCI}_{d,v} + \text{RSI}_{d,v} + \text{VEI}_{d,v} + \text{ACI}_{d,v}). \quad (2)$$

For matched-interface controls, each scored generation has

$$\text{AFCI}_{i,v} = \frac{1}{4} (R_{i,v} + M_{i,v} + Q_{i,v} + S_{i,v}), \quad (3)$$

where R , M , Q and S denote reason-action, memory-action, veto-action and self-action alignment terms. The implementation uses emitted trace fields, first impulse, final action and predefined field-source metadata to evaluate these terms. Thus APCI is an interface-level action-field coupling analysis.

For the finite-action-code benchmark, the canonicalizer maps the raw final action o_i to $\hat{y}_i = g(o_i) \in \mathcal{Y}$, where $\mathcal{Y} = \{\text{ACTION_A}, \text{ACTION_B}, \text{VETO}, \text{DEFER}, \text{RECALL_PRIOR}, \text{INVALID_OR_UNMAPPED}\}$. If y_i^- is the scorer-side expected action before the manipulation and y_i^+ is the scorer-side expected action after the manipulation, then

$$c_i = \mathbf{1}\{\hat{y}_i = y_i^+\}, \quad (4)$$

$$u_i = \mathbf{1}\{z_i = \text{irrelevant}\} \mathbf{1}\{\hat{y}_i \neq y_i^-\}. \quad (5)$$

The component score for component k is the target-minus-irrelevant contrast

$$B_k(v) = \widehat{\Pr}(c_i = 1 \mid z_i \in \mathcal{T}_k, v) - \widehat{\Pr}(u_i = 1 \mid z_i \in \mathcal{N}_k, v), \quad (6)$$

where $\mathcal{T}_k$ is the target-probe set and $\mathcal{N}_k$ is the irrelevant or placebo-probe set.

Dataset-cluster bootstrap intervals resample dataset identifiers with replacement, following the non-parametric bootstrap principle of estimating uncertainty from resampled empirical units (Efron and Tibshirani, 1993). For a metric m , comparator q , and dataset d , the paired contrast is

$$\delta_d(q; m) = \bar{m}_{d, \text{A4}} - \bar{m}_{d, q}, \quad (7)$$

with A4 replaced by A4_full for matched-interface APCI. The reported lower 95% bound is the empirical 2.5th percentile of the bootstrap distribution over $|\mathcal{D}|$ -dataset resamples.

For entropy calibration, layer-specific entropy $H_{d,v}^{(\ell)}$ is computed for raw, rule-canonical, semantic-cluster and action-family representations. If $\theta = (T, p)$ denotes a candidate high-stochasticity temperature/top-p setting, the predefined calibration gap is

$$\Gamma(\theta) = \max_{\ell} \left| \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} H_{d, \text{A5}(\theta)}^{(\ell)} - \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} H_{d, \text{A4}}^{(\ell)} \right|. \quad (8)$$

The predefined criterion requires $\Gamma(\theta) \leq 0.15$. The retained calibration attempts did not satisfy this criterion; the entropy results are reported as calibration diagnostics.

S9. Open-Weight and Formal Validation Controls

This section records the validation experiments that support the main-text evidence summary. The baseline, matched-interface, finite-action and entropy-diagnostic sections above provide the primary mechanistic evidence and calibration checks. The rows below summarize the open-weight finite-action matrix, strict-lesion slice, decisive-field positive-control experiment, local control-binding wrapper, formal control-audit criteria, local forced-budget entropy/token matching, entropy-prior behavioral control, the SWE-bench Lite issue-to-file localization validation, counterfactual-transform audit, API validation and expanded blinded annotation.

Table 33: Validation controls and interpretation status.

Evidence block	Scope	Main result	Interpretation status
Open-weight matrix	finite-action Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, Qwen2.5-32B-Instruct and Mistral-7B-Instruct-v0.3; 126,000 scored generations per model; three agent scaffolds; 20 task families; seven variants; three seeds	Structured accuracy exceeded distribution-matched offline/no-context controls in 236/240 model-scaffold-task cells, scrambled-context controls in 240/240 cells and no-fields controls in 238/240 cells; mean deltas were 0.593, 0.895 and 0.550, respectively	Supports open-weight size and family validation of the no-context control result
Independent family	Mistral-7B-Instruct-v0.3 full matrix	Core no-context controls satisfied the predefined criteria in 60/60 cells for no-fields, scrambled-context and distribution-matched controls	Supports non-Qwen model-family generalization
Strict Qwen32B target lesion	Qwen2.5-32B strict lesion slice; 28,800 scored generations	Structured accuracy exceeded strict target lesion in 60/60 cells; structured mean 0.998, strict-lesion mean 0.037, mean delta 0.961	Supports the narrower strict-lesion claim; ordinary target lesions remain a boundary condition
Decisive-field positive control	gpt-5.4-mini and qwen3-max ; 20 task families; six events per family; 1,440 fallback-free API generations	Full-state and only-decisive-field accuracy were both 1.000 in both models, while the best non-decisive control was 0.208; the only-decisive field is intentionally visible and action-code-bearing	Supports a diagnostic decisive-field readout within the finite-action protocol
Local control-binding wrapper	Same 1,440-row positive-control experiment; no second model call	Irrelevant-cue accuracy remained 1.000; raw scrambled-field following was 0.958 for gpt-5.4-mini and 0.533 for qwen3-max , reduced to 0.033 in both models after wrapping	Finite-action guardrail evidence; not real-environment localization or task-execution reliability
Formal Qwen control-audit slice	63,360 planned and observed rows; no duplicate analysis keys; parse/error rate 0.140%	Structured accuracy exceeded no-fields, scrambled-context, distribution-matched, strict-lesion and formal control variants in the relevant cells	Supports formal-control behavior in the Qwen slice
Token-prompt audit slice	17,280 planned and observed rows; no duplicate analysis keys; parse/error rate 0.069%	Prompt-token counts were aligned for the entropy-token variant, but mean total-token ratios and latency proxies remained outside the strict token/compute criterion	Report as token-prompt controls with compute audit, not strict token/compute matching
Strict entropy criterion	Formal entropy calibration and holdout audit	Registered entropy gap criterion was not met across model-scaffold cells	Strict entropy matching is not established
Local forced-budget entropy/token audit	Qwen2.5-1.5B-Instruct CPU local inference; 20 task families; two events per family; 80 generations	Prompt and completion tokens matched exactly by event; total-token ratio 1.000; latency ratio 0.997; canonical-action entropy gap 0.020 bits; structured accuracy was 0.900 versus 0.325 for the action-prior no-state control	Bounded local strict-matching pass; original formal strict entropy/token gates remain boundary conditions
Entropy-prior no-field behavioral control	Qwen2.5-14B and Qwen2.5-32B calibration and holdout rows; 24,480 planned and observed rows; no duplicate analysis keys; parse/error rate 0.102%	The entropy-prior no-field behavioral control remained below the structured-control variant in 120/120 cells, with mean delta 0.660; the strict entropy-gap criterion was satisfied in only 1/6 model-scaffold cells	Report as a final-action entropy-prior behavioral control, not strict entropy matching

Evidence block	Scope	Main result	Interpretation status
SWE-bench Lite issue-to-file validation	<code>princeton-nlp/SWE-bench_Lite</code> test split; 300 issue records; six API models; 18,000 condition/repeat rows aggregated to 1,800 model-issue records for the primary AUC analysis	Adding the causal state-binding diagnostic composite to a gold-free full non-CSB baseline increased task-only implementation-file hit@3 AUC from 0.732 to 0.915, with Δ AUC 0.183 and issue-cluster bootstrap lower 95% bound 0.095; leave-one-repository and leave-one-model sensitivities were positive in 12/12 and 6/6 analyses	Real issue-record localization predictor evidence; patch/test execution and issue resolution are outside this endpoint
SWE-bench Lite reliability calibration	Same 1,800 model-issue records; issue-grouped five-fold logistic probabilities for the gold-free full non-CSB baseline and baseline plus CSB	Reliability curves and threshold analyses stratify predicted task-only hit@3 probability, task-only hard-constraint violations and task-only constraint-clean hit@3 across acceptance thresholds	Secondary issue-to-file reliability analysis of the same endpoint
Counterfactual transform QC	Deterministic counterfactual-flip transform-quality screen	Text-label consistency screening retained 57/240 candidate counterfactual rows as transform-quality boundary rows	Interpret only consistency-passing transforms as robustness evidence
Consistency-passing counterfactual transform	Frozen transform on Qwen2.5-14B and Qwen2.5-32B with jointly regenerated decisive context and expected action	The consistency-passing transform satisfied the predefined criterion in 120/120 scaffold-family cells; mean structured accuracy was 1.000, compared with 0.992 in the original structured rows	Use as deterministic-transform robustness evidence
Expanded blinded annotation	1,200 full-overlap items, three annotators	Core dimensions had 100% valid scoring, no unscorable items and no violations of the blinding protocol; mean pairwise Cohen’s kappa was 0.961 and the lowest non-constant dimension kappa was 0.860	Supports annotation reliability
Formal API external validation	Provider-reported run-time identifiers <code>gpt-5.4-mini</code> , <code>gemini-3.1-flash-lite-preview</code> , <code>deepseek-v3.2</code> and <code>qwen3-max</code> ; 14,400 planned rows; 11,520 fallback-free remote API calls	All rows were present, duplicate analysis keys were zero, fallback calls were zero, unrecovered generations were zero and one parse/action error record remained. Structured accuracy exceeded no-fields, scrambled-context, distribution-matched and strict-lesion controls in 238/240, 240/240, 240/240 and 240/240 model-scaffold-task-family cells; all model-, scaffold- and task-family-cluster bootstrap lower 95% bounds were positive	Supports formal external validation of the core no-context and strict-lesion controls

The decisive-field positive-control experiment is a controlled finite-action test. The only-decisive condition exposes the current event’s minimal action-relevant field and asks whether the model follows that field rather than surface context, a calibrated action prior or a decisive field from another event. It is not interpreted as unconstrained task solving or hidden reasoning. The examples below show the prompt-level semantics used for the minimal field; the expected action column remains scorer-side metadata used for evaluation, whereas the visible field is the intervention-carrying state supplied to the action query.

Probe component	Example visible only-decisive field	Scorer-side target role
Reason	Decisive reason field for the event: calibrated task evidence selects <code>ACT_B</code> .	Target checks whether final action follows the current reason field.
Memory	Decisive memory field for the event: the corrected prior commitment is <code>ACT_A</code> ; stale or unrelated memory is non-decisive.	Target checks whether final action follows corrected memory rather than injected memory.
Veto/constraint	Decisive constraint/veto field for the event: a hard constraint invalidates ordinary actions and requires <code>ACT_VETO</code> .	Target checks whether final action respects the current constraint field.

The decisive-field leakage and scope audit separates hidden scorer metadata from intentionally visible decisive-field content. The only-decisive condition is not an answer-token-free hidden-reasoning task: it

deliberately exposes the current event’s minimal action-relevant field. The appropriate leakage question is therefore whether the scorer target variable or target key was serialized as metadata, and whether non-decisive controls withheld or broke the current decisive field.

Audit item	Evidence	Status	Interpretation
<code>expected_action</code> column not a prompt key	Prompt construction serializes condition, event id, task family, candidate actions and context; <code>expected_action</code> is retained in traces for scorer-side evaluation after generation.	pass	The scorer target variable is not exposed as a prompt field or JSON key.
Scorer-side target post-generation	used Across 1,440 traces, <code>expected_action_match</code> is computed after generation from canonical <code>final_action</code> versus <code>expected_action</code> .	pass	The target column is a scorer-side annotation rather than a model-output field.
Non-decisive controls do not expose current decisive state	do <code>surface_context_only</code> and <code>action_prior_only</code> had <code>field_event_id=none</code> in 240/240 rows each; <code>scrambled_decisive_field</code> used a non-current field id in 240/240 rows.	pass	Surface, prior and scrambled controls separate action priors or wrong-state fields from the current decisive field.
Only-decisive action-token status	<code>only_decisive_causal_field</code> exposed a visible decisive action matching the scorer target in 240/240 rows; full-state and irrelevant-cue rows did the same by design.	scope	The positive-control claim tests following the current minimal decisive field; it is not an answer-token-free hidden-reasoning claim.
Label wording scope	The prompt contains a generic instruction not to infer hidden expected-action labels, but does not serialize the <code>expected_action</code> column or scorer target key.	pass	The generic instruction names the evaluation concept; it does not provide the hidden scorer variable.

Scope of validation and execution slice. The formal criteria are interpreted conservatively. No-context and formal-control contrasts support the bounded claim that measured finite-action performance depended on structured state information in the tested agent protocol. Strict entropy and strict token/compute criteria that were not satisfied in the original formal slices are reported as limitations rather than post hoc passes. The local Qwen2.5-1.5B forced-budget audit is a separate bounded pass under exact prompt padding and fixed generation length. The SWE-bench Lite result is a full 300-record issue-to-file localization validation; patch application and test-suite execution were evaluated only in an exploratory slice. That frozen 50-record slice included official harness outcomes for two selected models, with 1/100 selected-model rows resolved and an inconclusive association gate. A later v3 apply-check/repair filter retained 8/100 locally applicable selected-model patches, eliminated harness errors on that filtered subset, and resolved 0/8. These outcomes define the scope of the localization result. The counterfactual-transform quality-control screen separates transform-quality rows from the consistency-passing deterministic transform used as robustness evidence.

References

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